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Joint optimization of electric bus scheduling and fast charging infrastructure location planning

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ABSTRACT

Transit authorities are transitioning from conventional buses to electric buses (EBs) due to growing concerns about air quality and greenhouse gas emissions. Many mathematical optimization models have been developed for scheduling conventional buses. However, such models would not fit EBs due to their limited travelling range and long charging time. Such operational differences have prompted new research into the literature on the charging station location problem. This study combines EB scheduling with fast-charging infrastructure location planning to minimize total scheduling and charger installation costs. We propose an Integer Linear Programming (ILP) formulation for a path-based model, solved using four developed branch-and-price algorithms, and test their performance across various instances. Our computational experiments show which branching strategy is computationally efficient in terms of execution time and optimality gap. Finally, we conduct a real case study and perform a sensitivity analysis to identify the most cost-effective type of electric bus, considering the specific characteristics of different EB types.

1. Introduction

Greenhouse gas-emitting energy sources are responsible for global warming. Thus, replacing such energy sources with clean and renewable sources has become crucial during the past few decades. Aligned with this, transit agencies are progressively transitioning towards electrifying their transit systems. They are increasingly embracing electric buses (EB) as an energy-efficient and emission-free alternative to conventional bus fleets. Thus, there has been a significant increase in the utilization of electric vehicles and electric buses. According to Mulholland and Rodríguez (2022), zero-emission buses are expected to account for approximately 32% of the bus fleet in European capital cities by 2030. The European electric bus market is projected to increase by 18.6% between 2022 and 2027, indicating a shift towards electric buses from conventional buses.

Transit authorities are increasingly focused on enhancing the efficiency of their public bus transit systems and one of the most important factors affecting that is operational processes (Kang and Meng, 2017). Therefore, optimizing bus operation planning has become essential to reduce operational costs and decrease passengers waiting time. Bus scheduling is one of the most vital processes

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in bus operations and has a considerable impact on operational costs. Bus/vehicle scheduling (VS) is the process of assigning buses to the trips of a given timetable and aiming to use the minimum number of buses while minimizing operating costs. EBs have battery capacity limitations and their scheduling and charging processes differ from those of conventional buses (Yao et al., 2020). The operating range of EBs is less than conventional buses and their charging time is longer than refueling diesel buses. These differences necessitate the development and adoption of new plans and strategies for scheduling EBs effectively. Li (2014) demonstrates how the introduction of electric buses affects the bus scheduling process, highlighting the impact of charging station capacity and charging time constraints.

To overcome the challenges of limited driving range and extended charging times, public transportation agencies are increasingly adopting fast-charging (FC) technology, which utilizes high-voltage electricity to recharge electric buses within minutes. This innovative solution has made fast-charging systems an increasingly popular choice for bus transit networks. Also, charging stations at bus terminals are less costly than depot charging and better suited to bus electrification throughout the life cycle (Lajunen, 2018). The limited travelling range of EBs has prompted new research into the literature, which is the problem of locating charging stations for electric buses. This problem identifies the ideal places for installing fast-charging infrastructures on the bus transportation network, as well as the optimal number of such stations to minimize the total establishment costs. However, using such technology makes EBs scheduling more complex (He et al., 2020). For example, the location of fast chargers affects the travelling time of charging trips of electric buses; hence, EB scheduling should be customized to consider this charging trip time. Also, the time of charging due to the limited availability and capacity of charging stations should be coordinated. Thus, as highlighted by Bie et al. (2021), a reasonable charging strategy is necessary to improve bus scheduling, and it is crucial to study the fast-charging infrastructure location planning problem for electric buses.

Transit authorities opting for fast-charging technologies employ pantograph chargers capable of delivering up to 350 kW of power. Although initially more costly, pantograph systems could ultimately decrease the total cost of ownership (TCO) and operational expenses for EBs as they allow continuous opportunity charging along routes (Alamatsaz et al., 2022). This reduces the need for large battery capacities and energy consumption. Moreover, pantograph charging enables an automatic connection from the bus roof, which can be integrated into existing bus networks by installing fast chargers at particular stops. This integration supports continuous bus operation with lower maintenance costs and higher availability, allowing buses to run around the clock (Al-Saadi et al., 2022). This mechanism is a well-established method for linking electric buses with their power sources. A special inverted pantograph automatically lowers when a bus arrives at the charging station. After completing the safety checks, the system provides the bus with a rapid recharge.

Transitioning from traditional buses to electric buses requires further research on public transit operation planning steps. These steps are network route design, timetable development, vehicle scheduling, and crew scheduling. First, the network route design outlines the bus routes and their bus stops based on the regions' geographical layout and travel demand. Next, the timetable development sets bus departure frequencies in order to cover the travel demand of passengers. In the vehicle scheduling step, buses will be efficiently assigned to bus routes based on the timetables. Finally, the last step determines the schedule for bus drivers. The limited range and longer charging time of EBs necessitate a reassessment of these steps. The majority of studies discuss public transportation operation planning in a sequential manner, where the output of one planning phase is used as input for the next stage. This approach is less effective compared to a fully integrated strategy. The fully integrated method analyzes the problem as a whole, considering all phases of public transportation planning simultaneously. For instance, making minor adjustments to the bus timetable might lead to better vehicle scheduling, and locating fast-charging infrastructure based on bus schedules could lower the operational costs of bus scheduling. Thus, in this research, we aim to integrate charging station location planning and bus scheduling steps of electric bus transit operation planning to improve the overall efficiency of the transit system (Olsen and Kliewer, 2022).

In the network design step, the travel demand of each region is analyzed to design the bus routes and their bus stop locations based on the geographical features and travel demand of that region. It also includes determining the length of bus routes ensuring the routes are of a length that EBs can complete their cycles without battery depletion. Finding the best spots to locate charging stations or continuous charging infrastructures is another aspect of the network design step for EBs. Also, this step includes choosing the most suitable charging technology based on the EBs' specifications and bus network features. The vehicle or bus scheduling step involves assigning the buses to each route based on the vehicle's range, charging requirements, given timetable, and route's length. The aim is to cover all the given trips in timetables with minimum operational costs. In this step, planners seek to minimize the total number of required EBs, total deadhead trips, and charging costs while covering all the given trips. Note that the charging scheduling of EBs is considered in this step.

In this research, we address the fast-charging location planning and electric bus scheduling problem (FCLP-EBS), which integrates strategic network design decisions with operational scheduling decisions. Given a set of bus trips, candidate charger locations, and fleet characteristics, the problem consists in determining: (i) the schedule of electric buses needed to cover all trips and (ii) the number and location of fast chargers to be installed on the network, such that operational constraints related to energy consumption, charging requirements, and trip coverage are satisfied. The objective is to minimize total operating costs, including deadhead travel, electricity costs, and electric bus ownership costs, as well as charger installation costs. To solve this problem, we introduce a path-based model (PBM) and apply a branch-and-price algorithm with multiple branching strategies.

The contributions of this study are as follows:

- We propose a new integrated FCLP-EBS model that more closely reflects real-world conditions by explicitly considering bus charging schedules and charging station capacities. This contrasts with recent studies (e.g., Wang et al., 2022; Davatgari et al., 2024) that assume fast chargers are always available.

- We develop a branch-and-price (B&P) algorithm to solve the FCLP-EBS problem. We propose an innovative network structure to reduce the number of network arcs and speed up the solution process.
- We compare the integrated approach with the sequential approach (fast charger location, followed by bus scheduling), show the advantages of the integrated approach, and demonstrate the model on a real case study in Montreal with sensitivity analyses across EB types.

We adopt an exact solution approach for small and medium-sized instances. For large-scale instances, we employ heuristic branch-and-price methods to obtain near-optimal solutions with an average optimality gap of less than 1%.

The remainder of this article is structured as follows. A literature review on charging station location planning for EBs, electric bus scheduling problems, and their integration is presented in Section 2. In Section 3, we provide a detailed description of the fast-charging location planning and electric bus scheduling problem. This section also describes the proposed path-based model. Section 4 provides an overview of the branch-and-price algorithm used to solve the model and the branching strategies. Then, Section 5 presents the computational results and includes a real-life case study to demonstrate the effectiveness and efficiency of the proposed solution approach. Finally, conclusions are drawn in Section 6.

2. Literature review

The following sections will provide a brief overview of the related works and theoretical background of charging station location planning and EB scheduling. Section 2.1 describes the different charging station location models found in the literature. In Section 2.2, we review the literature on EB scheduling problems, focusing on exact and heuristic methods used to address these problems. Finally, the integration of charging station location planning and EB scheduling problems are briefly discussed in Section 2.3.

2.1. Charging station location

Adopting fast chargers for electric buses to recharge them in a short time is becoming more common. Due to this growth, extensive research is required to determine the best locations for installing charging station infrastructure along bus routes. Kunith et al. (2014) conducted one of the initial studies on the planning of fast-charging stations. The study aimed to identify the optimal locations for fast chargers throughout the bus network to meet the daily energy demand of EBs. Csonka (2021) addressed the problem while considering partial charging, i.e., how much to charge EBs during each charging event. In another study, Olmos et al. (2019) looked into the problem of locating opportunity charging infrastructure for hybrid and fully electric buses to cover the required charging activities that fulfill the energy demand of EBs.

Alwesabi et al. (2021) provided a mixed-integer linear programming (MILP) formulation for determining the battery capacity and dynamic wireless charging locations. Tzamakos et al. (2023) investigated fast-wireless-charging infrastructure location planning, taking into account the charging scheduling. Considering the optimal capacity for EBs Kunith et al. (2017), the optimal number and types of EB chargers (Zhou et al., 2023), battery aging (Berthold et al., 2017), the power distribution network to reduce the power loss costs (Wu et al., 2021), and added energy storage system (ESS) to store the energy during off-peak hours and sending it to fast-charging infrastructure during peak hours (He et al., 2019) are other extensions of this problem. Hu et al. (2022) addressed the opportunity charging location problem while taking into account the charging scheduling of EBs. The authors aimed to minimize the cost of procuring opportunity chargers and EB batteries, as well as the total cost of charging and the extra passengers' waiting time by proposing a robust optimization model. The main distinction between their study and ours is that they neglect to consider the EB scheduling problem and the number of buses required to cover all trips. In addition, their model focuses on a single depot and does not incorporate Time-of-Use (TOU) electricity pricing and interlining of buses.

In earlier literature, most studies treat the energy demand for EBs as a constant. Furthermore, these studies assume that the bus schedule is predetermined. Consequently, the primary focus was identifying the best locations for charging stations and their optimum number, ensuring EBs can complete their trips based on fixed schedules. However, our research introduces flexibility by treating the bus schedule as a decision variable that can be adjusted according to the charging station locations and the number of such facilities. Thus, this research seeks to find the optimum solution for the fast-charging station location planning while optimizing the scheduling of electric buses. This integrated approach promises more effective solutions than the traditional sequential methods where the electric bus schedules are set in advance (see Section 5.2.2 for more details).

2.2. Electric bus scheduling

Vehicle scheduling (VS) is the process of assigning buses to the routes of a given timetable. Bus scheduling aims to cover the given timetable efficiently while satisfying all operational constraints by finding the necessary number of buses and reducing the deadhead trip costs. The number of possible solutions for EB scheduling is especially large when the EBs are based at multiple depots or terminals. The problem of electric bus scheduling across multi-depots falls within the category of NP-hard problems (Carpaneto et al., 1989). As a result, much of the research in this area focuses on computational challenges. Perumal et al. (2022) and Zhou et al. (2024) provided detailed literature reviews of the constraints associated with the electric bus scheduling problems and the methodologies proposed to address them. A well-known approach to model vehicle scheduling problems was introduced by Bertossi et al. (1987) and is known as the time-space network framework. In this framework, nodes represent the trips and their starting and ending times, while arcs represent the routes taken by vehicles to get from one node to another. We use this framework to model the integration of EB scheduling and fast charging location planning problem.

The literature on electric bus scheduling problems offers a wide variety of models and methods. Yao et al. (2020) solved the EB scheduling problem using a Genetic Algorithm (GA) for scheduling heterogeneous types of electric buses. Teng et al. (2020) integrated bus timetabling with EB scheduling and aimed to optimize the overall efficiency of the bus line, reducing operational costs while ensuring timely service. Zhou et al. (2020) integrated the EB scheduling problem with the charging management of electric buses by proposing a bi-level programming approach. The model simultaneously addresses vehicle scheduling and charging management, optimizing the overall operational efficiency of the fleet. Similarly, Zhou et al. (2025) considered the impact of EB charging schedules on the power grid within the context of bus scheduling. They employed the NSGA-II algorithm, a multi-objective evolutionary approach, to solve the problem. Sung et al. (2022) proposed a heuristic solution approach to tackle the EB scheduling problem by considering a mix of heterogeneous buses and chargers for multi-depots and multi-bus line networks. Wang et al. (2024) addressed the same problem while considering restricted charging facilities. The authors proposed an adaptive large neighborhood search (ALNS) algorithm that incorporates innovative acceleration techniques to handle large-scale instances efficiently. Their proposed method was validated through a case study in Lianyungang, China. As these studies rely on heuristic or metaheuristic methods to solve the electric bus scheduling problem, they do not guarantee optimal solutions.

On the other hand, there are a few studies focusing on exact solution approaches to deal with EB scheduling problems. Notably, Li et al. (2019) developed a scheduling model for mixed bus fleets, addressing the challenges of electric buses' range limitations and refueling needs. They have adopted a time-space-energy network approach and developed an integer linear programming model to solve the problem. Gkiotsalitis et al. (2023) presented an exact approach based on a branch-and-cut (B&C) algorithm for the multi-depot electric bus scheduling problem, to address operational constraints like time windows and charging requirements. Tang et al. (2019) addressed the single-depot problem by robust scheduling strategies for electric buses, focusing on the uncertainties introduced by stochastic traffic conditions to minimize total operating costs. The authors used a branch-and-price method to tackle the problem. Bie et al. (2021) proposed an optimal scheduling strategy for electric buses by combining all-stop and short-turning methods. Their model aims to balance the operational efficiency of electric buses with energy consumption constraints, enhancing overall system performance for a single bus line. Jiang and Zhang (2022) developed an advanced branch-and-price algorithm to solve the large-scale multi-depot electric bus scheduling problem, which is more complicated compared to the single-depot problem. Lu et al. (2025) proposed a mixed-integer programming (MIP) model that integrates both vehicle assignment and charging decisions while considering the constraint of TOU pricing that penalize peak-hour charging. The authors solve the problem by using a branch-and-price algorithm and benchmark against heuristic methods.

A nonlinear optimization model is proposed in the study of Zhang et al. (2024) to address the electric bus scheduling problem while accounting for constraints like limited charging stations and power availability. The approach combines bus scheduling with the optimization of charging times and power usage to minimize total system costs, including vehicle operation and charging expenses. A genetic algorithm was used to solve the model. Similarly, van Kooten Niekerk et al. (2017) presented two methods for scheduling EBs with limited travel ranges. The first method uses a simplified model assuming linear charging regardless of time-of-use, while the second adjusts this to enhance realism. Both models were solved for small to medium-sized problems using integer linear programming. For larger-scale instances, the authors proposed two near-optimal solutions using the column generation method. In another study by Zhang et al. (2022), the authors seek to optimize the use of EBs by allowing flexible routing and partial recharging strategies, which helps to overcome limitations in battery range and charging infrastructure. Further advancing the field, Wu et al. (2022) addressed the multi-depot electric vehicle scheduling problem, incorporating power grid characteristics into the scheduling process. Additionally, electric bus scheduling under energy consumption and travel time uncertainty have been studied by Avishan et al. (2023).

2.3. Integrated fast-charging station location and bus scheduling

Recently, there has been a growing trend in integrating charging station location planning with electric bus scheduling. Olsen and Kliewer (2022) studied the combination of depot charging location planning with EB scheduling for a single-depot setup. The goal was to reduce the total cost, which included operational costs, installation costs of chargers in the depot, and vehicle costs. To address this problem, a metaheuristic solution method based on Variable Neighborhood Search was proposed. They proved that optimizing these two problems simultaneously outperforms the sequential planning approach. However, their study did not develop an exact solution method to tackle the problem and did not account for time-of-use electricity pricing. Li et al. (2019) explored the multidepot and mixed fleet version of this problem. For small-scale instances, they proposed an arc-based model and formulated it as an integer linear program, while for large-scale instances, they introduced a time-space bus flow network to find approximate solutions. They did not account for deadhead trip costs, partial charging, and time-of-use electricity pricing. Additionally, the maximum number of trips they solved was limited to 288, which is significantly lower than in our study.

Li et al. (2020) presented a joint optimization model for EBs, addressing both the scheduling of EBs and the deployment of chargers. The study incorporates a partial charging policy and time-of-use electricity prices to balance operational efficiency with energy costs. By optimizing the timing of bus charging and strategically placing chargers using an adaptive genetic algorithm (AGA), the model aims to minimize total operational and electricity expenses. While the model demonstrates the capability to handle up to 1260 trips, the authors do not report key performance metrics such as the optimality gap or solution runtime, leaving the quality of the solutions unclear. In contrast, our research proposes an exact solution approach that guarantees optimal solutions. Additionally, the study focuses on depot-based charger deployment, whereas we address the more operationally challenging case of fast chargers. Fast chargers significantly reduce charging time but require more frequent detours to charging stations due to the limited travel range of EBs. This added complexity increases the difficulty of managing charging schedules and routing, which is a key focus of our work.

Stumpe et al. (2021) investigated the simultaneous optimization of opportunity charging location planning and electric bus scheduling. They suggested an innovative mixed-integer linear formulation and used Variable Neighborhood Search metaheuristic to solve it. However, they considered single-depot problem and neglected to consider multiple depots for scheduling EBs. Also, opportunity charging is different from fast charging pantograph chargers. The pantograph chargers are usually located at the beginning or end of bus lines and their total number is typically less than opportunity chargers that are located at specific bus stops of bus lines. The charging duration of pantograph chargers, 5 to 15 min, is longer than that of opportunity chargers, which is only a couple of minutes. Also, the charging with opportunity chargers can occur when passengers are on the board, but for pantograph chargers, the EBs should be emptied. The opportunity chargers of each bus line are typically for the buses running that bus line. However, pantograph chargers are shared with the EBs running different bus lines. That is why the charging scheduling of pantograph chargers is more challenging and requires a different formulation and methodology to deal with. In this study, one aspect is to address the charging scheduling of EBs with pantograph chargers to smooth the charging events for EBs with the minimum waiting time of EBs at the charging stations. An additional advantage of our research is that we use an exact solution approach to achieve the optimal solution, whereas (Stumpe et al., 2021) employed a metaheuristic approach, which does not guarantee to find the global optimum.

One of the closest works to our research is the work of Liu and Ceder (2020). They proposed a bi-objective integer programming method for the problem with the objective of reducing the number of EBs and fast-charging infrastructures. The authors proposed two approaches to address this problem: a lexicographic approach and a modified max-flow technique. However, their study does not consider time-of-use electricity pricing, which is critical for realistic and cost-effective EB scheduling. Furthermore, the scale of their model, in terms of the number of trips and candidate charger locations, is significantly smaller than that addressed in our research.

Wang et al. (2022) optimized the deployment of pantograph chargers for electric buses while also optimizing the scheduling of buses. They developed a mixed-integer linear programming model to minimize overall fleet costs. However, their work did not address multi-depot operations, partial charging, travel costs, charging costs, and the scheduling of EB charging. Additionally, their results, obtained using a commercial solver, report an optimality gap of 4% for a case study with only 100 trips. In contrast, our study adopts an exact solution method for small to medium-sized problems to ensure global optimal solutions. For larger instances, we incorporate heuristic branching strategies, enabling us to handle up to 1000 trips while maintaining an optimality gap of less than 1%, demonstrating both scalability and solution quality.

Davatgari et al. (2024) addressed a similar problem by developing a mixed-integer programming model that determines the optimal assignment of EBs to bus lines while accounting for battery constraints and identifying the best locations for charging station. Their approach significantly enhances operational efficiency and charger utilization by reducing the total number of chargers needed and minimizing vehicle idle time. However, the scalability of their model is limited as it can only handle instances with up to 60 trips, making it unsuitable for larger real-world transit networks. Additionally, their model does not account for several key operational features, such as multi-depot setups, TOU electricity pricing, charging scheduling, and interlining, all of which are addressed in the current research. Zhang et al. (2025) introduced a mixed-integer nonlinear programming model that addresses the same problem, focusing on key operational challenges such as battery capacity degradation and scheduling uncertainties. Their goal was to minimize total operational costs, and they demonstrated the economic and operational benefits of en-route charging through a real-world case study. However, they did not consider partial charging for EBs. The primary distinction between their research and ours lies in scope. While their experiments were limited to 84 trips, a single depot, and a single bus line, our model extends to multiple depots and multiple bus lines, offering a more realistic and comprehensive solution for large-scale transit systems. This broader scope improves the applicability of the model, but also introduces significantly greater computational complexity, which our solution methodology is designed to handle effectively.

Another relevant study is the work by Najafi et al. (2025), which tackles a similar problem while incorporating renewable energy resources into the model. Their objectives include minimizing the total cost of purchasing and maintaining EBs, chargers, and renewable energy systems, as well as reducing time-of-use electricity costs. However, several practical considerations are missing from their model. Notably, they neglect to address the multi-depot setting, which is essential for capturing the operational complexity of real-world transit systems. Their model is also limited to only three bus routes, whereas depot operators typically manage a much larger network of bus lines. In addition, travel costs and waiting time costs are not included, which are key components in accurately assessing fleet performance. Most importantly, the model does not allow for partial charging of EBs, a realistic and often necessary strategy in high-demand operations. In contrast, our research addresses all these critical factors, providing a more comprehensive and practical solution for electric bus scheduling and infrastructure planning.

Although electric bus scheduling, charging scheduling, and charging station location problems have been the subject of various studies, there is still more work to be done to bridge the gap between theory and practice. A review of the literature reveals that few studies have explored the influence of electric bus scheduling and charging scheduling on the location of fast-charging infrastructure and vice versa. However, these studies did not consider the aggregation of practical factors such as multi-depots, time-of-use (TOU) electricity pricing, interlining, and partial charging of electric buses with pantograph chargers, factors that we address in this study. In addition, the model explicitly optimizes the charging schedule of electric buses, preventing long queues at charging stations. Also, a review of the literature reveals that most existing studies overlook the use of exact solution methods and struggle to address large-scale instances effectively. The absence of exact solution methods in the literature limits the ability to benchmark heuristic approaches effectively and restricts insights into the true potential of optimal scheduling and infrastructure design. Additionally, the inability of the current methods to scale effectively to real-world scenarios highlights the urgent need for innovative strategies that can offer a balance between computational efficiency and solution quality. To address these shortcomings, we propose the use of a branch-and-price algorithm to solve the FCLP-EBS problem. This exact solution approach not only guarantees optimality for small and medium-sized instances but also provides a robust framework to address the complexity of real-world transit systems. Compared

with the literature, specifically (Jiang and Zhang, 2022) and Wu et al. (2022), whose largest solvable instances were limited to 400 trips, our approach scales to instances with up to 1000 trips, which enables us to solve a real-world case study.

3. Problem description and mathematical formulation

This section presents a detailed explanation of the fast-charging location planning and electric bus scheduling (FCLP-EBS) problem. Section 3.1 describes and discusses the problem, and Section 3.2 introduces the corresponding path-based model.

3.1. Problem definition

This paper studies the *Fast-Charging Location and Electric Bus Scheduling* (FCLP-EBS) problem, which integrates infrastructure planning decisions with the daily scheduling of electric buses. The problem jointly determines (i) the locations of fast-charging stations selected from a predefined set of candidate sites and (ii) the assignment of a homogeneous fleet of electric buses to a fixed timetable of trips. The objective is to minimize total system cost, including fleet size, deadhead travel, waiting time, electricity consumption, and fast-charging station installation costs.

The transit system consists of a set of depots O, a set of scheduled trips S, and a set of candidate fast-charging locations B. Each trip $i \in S$ corresponds to a single journey with passengers and is characterized by a fixed start time a_i, end time e_i, travel distance $\tilde{d}_i$, and travel time $\tilde{\tau}_i$, all specified by the timetable. A route operating at regular frequency therefore generates a sequence of distinct trips, each of which must be covered exactly once.

Each electric bus starts its daily operation at a depot, returns to the same depot at the end of the day, and can be assigned to at most one trip at any point in time. The number of buses dispatched from each depot is a decision variable. Movements of buses without passengers, required to reposition vehicles between trips, depots, and fast-charging stations, are referred to as *deadhead trips*.

Depots, fast-charging station locations, and the start and end points of trips are collectively referred to as *nodes*. Let d_{ji} and τ_{ji} denote the distance and travel time, respectively, of a deadhead movement from node j to node i. The set of all deadhead movements is denoted by $\hat{D}$, and ω_{ji} represents any waiting time incurred between arriving at node j and starting service at node i. Electric buses are subject to limited driving range. Let $\hat{d}_{\max}$ denote the nominal driving range of a fully charged bus and let $SOC_{\min}$ be the minimum allowable state of charge, expressed in distance units. The maximum effective driving range is therefore $d_{\max} = \hat{d}_{\max} - SOC_{\min}$. A bus may start a trip only if it has sufficient energy to complete the trip and subsequently reach either another assigned trip or a fast-charging station. After completing a trip, an EB may deadhead to the next trip (with potential waiting time) or travel to a fast-charging station for recharging.

The model allows *interlining*, meaning a single electric bus can cover multiple trips, even across different routes, without returning to the depot between trips. Starting fully charged, an electric bus can operate a sequence of trips that are compatible in time and location, as long as it stays within its battery limits and can access a fast-charging station when needed. Interlining improves vehicle utilization and reduces deadhead travel, but it also increases reliance on fast-charging station availability.

To model charging capacity constraints, the planning horizon is discretized into a set of time periods $T = \{1, \dots, \max_T/\Delta T\}$, each of length ΔT. Here, $\max_T$ represents the maximum available time to charge EBs, and $\max_T/\Delta T$ is assumed to be an integer. A charging event lasts for one or more consecutive full time periods. At each fast-charging station, at most W buses may be charged simultaneously. The parameter W specifies the station's charging capacity, i.e., the maximum number of buses that may be charged in parallel. The cost parameter V_b denotes the installation cost of opening a fast-charging station at location $b \in B$ and is independent of the value of W. The duration of any single charging event is limited to at most U minutes, which is the time required for a bus to fully recharge. To simplify the notation, we assume U is a multiple of ΔT.

Charging is assumed to follow a linear process using fast pantograph chargers. This assumption is appropriate for strategic and tactical planning. Strong nonlinearities in battery charging typically occur at very low and very high states of charge, whereas transit agencies usually operate buses within a mid-range state of charge to preserve battery health and ensure service reliability. In this operating range, a linear approximation of the charging process is reasonable. Moreover, the primary focus of this study is on fleet sizing, trip scheduling, and fast-charging station location. Explicitly modeling nonlinear, state-of-charge-dependent charging dynamics would significantly increase model complexity and reduce the size of solvable instances, with limited benefit for long-term planning decisions. This modeling choice is consistent with prior work in the electric bus planning literature (e.g. Mortensen et al., 2023; Bagherinezhad et al., 2020; He et al., 2020; Bao et al., 2023; Zeng et al., 2022; Liu et al., 2021).

The objective function includes several cost components. Operational costs consist of a distance-based cost C_d per kilometer traveled and a waiting cost C_w per minute. Each electric bus incurs a fixed annual cost C_v. Charging costs depend on time-of-use electricity pricing. Let C_t denote the cost per minute of charging during time period $t \in T$, with distinct peak and off-peak rates. By scheduling charging during off-peak periods when possible, total electricity costs and stress on the electrical grid can be reduced. Table 1 summarizes the main sets, parameters, and decision variables used in the formulation of the FCLP-EBS model.

Fig. 1 presents a schematic solution for an illustrative instance with two depots, five bus lines, and five candidate charging locations. Bus routes are shown in different colors, candidate fast-charging stations are indicated by charger icons, and deadhead movements are distinguished graphically: dash–dot arrows represent movements between depots and routes, while dashed arrows represent movements to and from fast-charging stations.

Table 1
Notation used in this paper.

Notation	Description	Notation	Description
Sets			
O	Set of depots	S	Set of trips
B	Candidate locations for fast-charging stations	T	Set of time periods
P	All possible paths from a depot and back	G	Network
N	Set of nodes	A	Set of arcs
R	Set of resources	$\hat{D}$	Set of all deadheads
$\bar{P}$	Subset of all feasible schedules		
Parameters			
C_v	Annualized cost of using a new bus	V_b	Annual cost of installing a fast charger
C_d	Unit travel cost per km	C_w	Cost per minute of waiting
C_t	Charging cost per minute at time $t \in T$	$\bar{c}_{ji}$	Reduced cost of arc (j,i)
c_p	Cost of path p	W	Capacity of a fast-charging station
U	Maximum charging time	$d_{\max}$	Maximum EB travel range
a_i	Start time of trip $i \in S$	e_i	End time of trip $i \in S$
d_{ji}	Deadhead distance $j \to i$	$\tilde{d}_i$	Distance of trip $i \in S$
τ_{ji}	Travel time $j \to i$	$\tilde{\tau}_i$	Travel time of trip $i \in S$
w_{ji}	Waiting time between nodes j and i	max_T	Latest time period
t^e_{ib}	Earliest EB arrival at charger b after trip i	$[l^r_i, u^r_i]$	Resource bounds at node i
ξ	Remaining electric range of an EB	w_p	Total waiting time in path p
ϖ^p_{ji}	1 if deadhead (j,i) is in path p; 0 otherwise	δ^p_i	1 if trip i is in path p; 0 otherwise
σ^p_{tb}	1 if path p charges at station b during time $t \in T$; 0 otherwise		
Decision variables			
y_p	1 if path p is selected; 0 otherwise	z_b	1 if a fast-charging station has been installed at location b; 0 otherwise

3.2. Path-based model (PBM)

We introduce a path-based model that features a large number of variables but a reasonable number of constraints. This structure allows the optimization model to effectively utilize the branch-and-price algorithm. We denote P as the set of all possible paths starting from the depots and returning to the same depots. Such paths or schedules should satisfy the maximum distance constraint prior to battery charging. Each path $p \in P$ represents a schedule in which an EB leaves a depot, operates several bus trips, with potential recharging event at fast-charging stations between two trips, and finally returns to its originating depot. Let $\varpi^p_{ji} = 1$ if deadhead (j,i) is included in path p. Similarly, let $\delta^p_i = 1$ if trip i is covered by path p. Let $\sigma^p_{tb} = 1$ if path p includes charging during time period $t \in T$ at fast-charging station b. Let w_p represent the total waiting time in path p, which includes the waiting time between trips, trips to/from chargers, and waiting time at the fast-charging stations. The cost of path p, denoted as c_p, is then given by:

$$c_p = C_v + \sum_{(j,i)\in\hat{D}} \varpi^p_{ji}\left(C_d d_{ji}\right) + \sum_{i\in S} \delta^p_i\left(C_d \tilde{d}_i\right) + C_w w_p + \sum_{b\in B}\sum_{t\in T} \sigma^p_{tb}\Delta T C_t \tag{1}$$

Let y_p be a binary variable equal to 1 if path p is selected, and 0 otherwise. Let z_b be a binary decision variable with $z_b = 1$ if a fast charger is installed at location $b \in B$, and $z_b = 0$ otherwise. The path-based formulation of FCLP-EBS problem is:

$$\min Z = \sum_{p\in P} c_p y_p + \sum_{b\in B} V_b z_b \tag{2}$$

s.t.

$$\sum_{p\in P} \delta^p_i y_p = 1 \quad \forall i \in S \tag{3}$$

$$\sum_{p\in P} \sigma^p_{tb} y_p \le W z_b \quad \forall b \in B, \forall t \in T \tag{4}$$

$$y_p \in \{0,1\} \quad \forall p \in P \tag{5}$$

$$z_b \in \{0,1\} \quad \forall b \in B \tag{6}$$

Objective function (2) minimizes the total annual operational costs of EBs and the total annual establishment costs of fast chargers. The first term is the total annual purchase and operational costs of EBs. The second term is the annual cost of establishing fast chargers

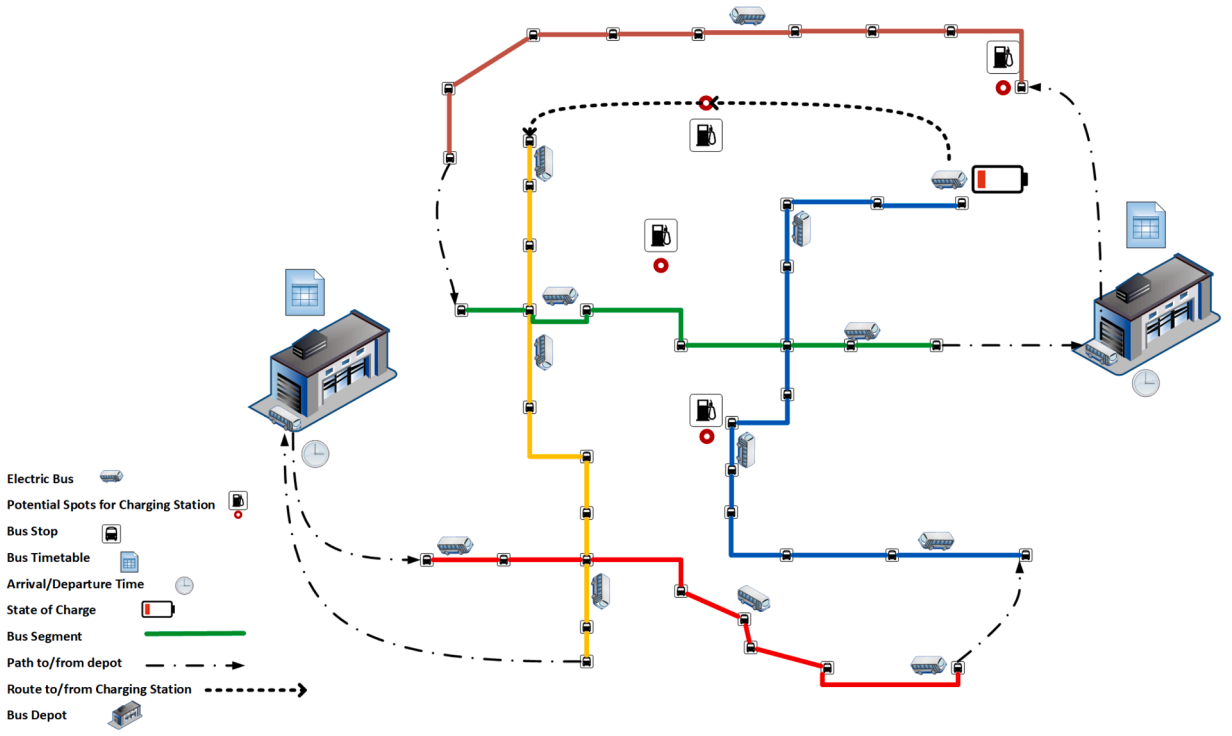


Fig. 1. A schematic view of a solution to the FCLP-EBS problem for an illustrative example with two depots, five bus lines, and five candidate locations for installing fast-charging stations.

along the network. Constraint (3) ensures that each trip is served once. Constraint (4) ensures that buses can charge only at locations where chargers are installed and that station-capacity limits are respected. Constraints (5) and (6) determine the domain of variables.

4. Methodology

Because the path-based formulation introduced in the previous section involves a very large number of variables, enumerating them all would be impractical. To address this, we solve the path-based model using a branch-and-price algorithm (B&P). B&P embeds a column generation algorithm inside a branch-and-bound tree. A linear relaxation of (2)–(6) is solved at each branching node, using column generation, to compute a lower bound (in the case of minimization). The column generation algorithm iteratively solves a *restricted master problem* (RMP) and one or several *pricing subproblems*. In our case, the RMP is the linear relaxation of (2)–(6) with the set of paths P replaced by $P' \subseteq P$. The goal of the pricing subproblems is to find a negative reduced cost path, which is then added to P'. The column generation algorithm iterates between solving the RMP and the pricing subproblems until no negative reduced cost path is found. At this point, the current solution is optimal for the linear relaxation of the problem. At the beginning of the B&P algorithm, P' is initialized with a set of artificial variables with a very high cost.

The following subsections detail each component of the B&P algorithm implemented. Section 4.1 presents the pricing subproblem, formulated as a constrained shortest path problem. It also describes the network structure and the labelling algorithm used to solve the subproblem. Several exact and heuristic branching strategies are evaluated, as presented in Section 4.2.

4.1. Pricing subproblem

Let $\lambda^{(3)}$ and $\lambda^{(4)}$ be the dual variable arrays for constraints (3) and (4), respectively. The reduced cost of path variable y_p, $p \in P$, denoted $\bar{c}_p$ is given by:

$$\bar{c}_p \equiv c_p - \sum_{i \in S} \delta_i^r \lambda_i^{(3)} - \sum_b \sum_t \sigma_{tb}^p \lambda_{bt}^{(4)} \tag{7}$$

The pricing subproblem is formulated as a constrained shortest-path problem in an acyclic network. Each constraint-feasible path in the network corresponds to a valid bus path and hence, a path variable in formulation (2)–(6). Furthermore, the cost of such a path is equal to the reduced cost of the corresponding variable. Path constraints are enforced by the structure of the network and by resource constraints, which are explained below.

4.1.1. Network structure

We first present the network structure for the single depot case and then expand it to the multi-depot case. Let $G = (N, A)$ represent the network, which contains five types of nodes and five types of arcs. A *route start* and *route end* node serve as source and sink, respectively. Each trip in S is represented by one *trip node*. Each fast-charging station is represented by two sets of time-expanded nodes. There is a *pre-charge* node for each fast-charging station/time period pair and a *post-charge* node for each fast-charging station/trip pair.

For each trip $s \in S$, there is a *beginning* arc between the route start node and trip node s. Likewise, an *ending* arc links trip node s to the route end node. *Deadhead* arcs connect trip nodes to one another, trip nodes to pre-charge nodes, and post-charge nodes to trip nodes. Specifically, there is a deadhead arc between trip node $i \in S$ and trip node $j \in S$ if $e_i + \tau_{ij} \leq a_j$ (i.e. it is possible to operate them both consecutively). For each fast-charging station $b \in B$ and each trip $i \in S$, there is a deadhead arc between trip node i and the pre-charge node of station b that corresponds to the earliest time period a bus can reach station b after ending trip i. Finally, for each station $b \in B$ and each trip $i \in S$, there is a deadhead arc between the post-charge node of station b and trip i.

We now describe the fast-charging station component shown in Fig. 2. At each fast-charging station $b \in B$, the internal network consists of two time-indexed node types: *pre-charge nodes* and *post-charge nodes*. Pre-charge nodes represent possible arrival times at the station prior to charging and are connected by *waiting arcs*, allowing buses to arrive early and wait. Post-charge nodes represent feasible departure times after charging and are also linked by waiting arcs. Their timing is defined so that a bus can still reach its next assigned trip on time; for example, for trip $i \in S$, the corresponding post-charge node at station b is associated with time $a_i - \tau_{bi}$. *Charging arcs* connect pre-charge nodes to post-charge nodes and represent charging activities. From each pre-charge node at time t, up to $U/\Delta T$ charging arcs are available, each corresponding to a different charging duration. For a charging arc of k time periods, the arrival node is the earliest post-charge node that allows the bus to feasibly proceed to its next trip. This structure allows a bus to arrive, wait, charge, wait if necessary, and then continue to serve subsequent trips while respecting time and capacity constraints.

Fig. 3 illustrates the internal operations of a fast-charging station. Suppose an EB arrives at trip node 7 with an insufficient remaining driving range and therefore must travel to charging station 1. The total time available for charging is 60 min. Since each time period is 5 min long, this results in 12 possible charging periods. Based on the ending time of trip 7, which is 8 in this example, and the deadhead travel time to charging station 1, a deadhead arc is created from trip node 7 to the earliest feasible time-expanded node at charging station 1. Note that only one arc is created from each trip node, linking it to the earliest available charging time. Therefore, in this case, trip 7 is not connected to any other time-expanded nodes. The model then determines whether the EB should wait or start charging immediately. In this example, the EB begins charging right away and charges for one time period, which is 5 min. Assume that $U = 20$ min and $d_{\max} = 100$. Under these conditions, the EB's driving range increases by 25 km. After the charging event, the EB proceeds to trip node 8.

4.1.2. Resources and cost computation

Path restrictions that cannot be enforced directly through the network structure are handled using *resources*. A resource is a commodity that is consumed along arcs. Bounds on resource consumption may be imposed at nodes; these bounds are referred to as *resource windows*. A path may enter a node only if its accumulated resource consumption satisfies the node's resource windows. The lower bound of each resource window is *soft*, meaning that a node may be visited even if the arriving resource level is below the lower bound. In such cases, the resource level is updated to the lower bound upon arrival. Let R be the set of resources. For each node $i \in N$ and each resource $r \in R$, the corresponding resource window is denoted by $[l_i^r, u_i^r]$, where l_i^r and u_i^r represent the lower and upper bounds, respectively, on resource r at node i.

Two resources are necessary in the pricing subproblems, which are *time* and *range*. The time resource keeps track of the time corresponding to a given partial schedule (in time period units). Each arc consumes time according to the travelling or waiting time it represents. The time resource windows for the source and sink nodes are respectively $[0, 0]$ and $[0, |T|]$. The time resource window at a trip node $i \in S$ is set to $[e_i, e_i]$ and the time resource window for post-charging node associated with i is $[0, |T|]$. Finally, the time resource window for pre-charge node corresponding to time period t is $[t, t]$.

Although an electric bus may arrive at the physical stop earlier than the trip start time a_i or may wait after completing the trip at time e_i, these waiting activities are explicitly modeled through waiting arcs in the time-expanded network. A bus that reaches the stop before a_i must follow waiting arcs until time a_i, at which point the trip can begin. Likewise, any waiting after completing the trip is represented through waiting arcs emanating from the trip node. Since Gencol operates using resource windows rather than fixed time values, the end time of each trip is defined as the interval $[e_i, e_i]$. Also, since the time resource records the completion time of the activities encoded in the network, the arrival time at the trip node must be exactly the scheduled end time e_i. The window $[e_i, e_i]$ therefore enforces the fixed timetable, while waiting flexibility is captured entirely through the structure of the network rather than through the node's time bounds.

The range resource tracks the accumulated distance travelled by an electric bus since its most recent charging event. We define the resource such that its value increases when the bus travels on a deadhead or trip arc, and decreases when the bus charges. Because of the soft lower bound applied to all resource windows, the lower bound is set to 0, which represents a fully charged battery (i.e., zero distance consumed). The upper bound at all nodes except the source is $d_{\max}$, the maximum allowable driving distance between two consecutive recharges. A path may only enter a node if its accumulated range consumption does not exceed $d_{\max}$. Thus, the upper bound ensures feasibility with respect to battery limitations: any partial path whose required energy exceeds the battery capacity is automatically pruned by the label extension rules. Charging arcs reduce the accumulated range consumption in proportion to their duration, which allows the model to represent partial charging events. Because nodes themselves do not enforce minimum distance requirements, the feasibility with respect to range is governed entirely by (i) the resource consumption associated with arcs, and (ii)

the upper bound $d_{\max}$ at each node. This design ensures that only paths satisfying all battery constraints are retained by the pricing subproblem, while still allowing flexible charging durations through appropriate selection of charging arcs in the time-expanded network.

The range resource keeps track of the remaining range for the EB. Because of the soft lower bound, the value of this resource is defined as $d_{\max} - \xi$, where ξ is the actual range remaining. The range resource is thus initialized at 0, representing the maximum range. It increases when the EB travels, and decreases when the EB recharges. Note that this inversion of the resource definition is necessary : otherwise, paths that violate the range constraints would be allowed due to the soft lower bound on the resource windows (the value would become negative, but increased to 0). The range resource window is $[0, 0]$ for the source and $[0, d_{\max}]$ for all other nodes. On each arc (i, j) where the bus travels (beginning, ending, and deadhead arcs), the resource consumption is d_{ij}. For a charging arc corresponding to charging u' minutes, the resource consumption is set as $-d_{max}u'/U$ (the range gained by the recharging activity). No range resource is consumed on waiting arcs.

Each arc (i, j) has a cost, denoted $\bar{c}_{ji}$, such that sum of the arc costs of a path is the reduced cost of the associated schedule variable. The cost of an arc includes the costs related to activities, as well as dual variables. The cost for each type of arc is listed below, with w_{ij} indicating the waiting time incurred while traversing arc (i, j).

- Beginning arc between depot o and trip node $i \in S$
  $\bar{c}_{oi} = C_d d_{oi} + C_d \tilde{d}_i + C_v - \lambda_i^{(3)}$
- Ending arc between trip node $i \in S$ and the depot o
  $\bar{c}_{jo} = C_d d_{jo}$
- Deadhead arc between trip node $i \in S$ and trip node $j \in S$
  $\bar{c}_{ij} = C_d d_{ij} + C_d \tilde{d}_i + C_w w_{ij} - \lambda_j^{(3)}$
- Deadhead arc between trip node $i \in S$ and pre-charge node for station $b \in B$ and time period $t \in T$ (denoted t^b)
  $\bar{c}_{itb} = C_d d_{ib} + C_w w_{it^b}$
  Note that waiting may occur only if the bus arrives at b between two time periods.
- Deadhead arc between post-charge node for station $b \in B$ and trip $i \in S$ (denoted i^b), and trip node i
  $\bar{c}_{i^b i} = C_d d_{bi} + C_d d_i - \lambda_i^{(3)}$
- Charging arc between pre-charge node for station $b \in B$ and time period $t \in T$ (denoted t^b) and post-charge node for station b and trip i (denoted i^b), corresponding to a charging event of u' time periods
  $\bar{c}_{t^b i} = \Delta T \sum_{t'=t}^{t+u'-1} C_{t'} + C_w w_{t^b i^b} - \sum_{t'=t}^{t+u'-1} \lambda_{bt'}^{(4)}$
- Waiting arc (i, j) between two consecutive pre-charge or post-charge nodes
  $\bar{c}_{ij} = C_w w_{ij}$

In the multi-depot case, the dummy arcs have no cost and consume no resources. It is easy to show that the cost of any resource-feasible path in the network corresponds to a feasible schedule variable and that the cost of this path is equal to the reduced cost of the associated variable.

The network for the single-depot case is presented in Fig. 2. An example of valid path is depicted in blue in Fig. 2. It represents a bus starting from the depot, performing trips 1, 5, potentially several others, then trip i. After trip i, the bus goes to fast-charging station and arrives at time t. It immediately charges for some periods, and then performs trip j, potentially other trips, and finishes with trip 9. The subproblem for the multi-depot case is represented in Fig. 4. A copy of the single-depot network is created for each depot. A new source node is created and linked to each route start node through dummy arcs. Likewise, a sink node is created and each route end node is connected to the sink via a dummy arc.

4.1.3. Solution method for the pricing subproblem

The pricing subproblem is solved using a standard label-setting algorithm, similar to the one described by Irnich and Desaulniers (2005). Although the full details of the algorithm are beyond the scope of this paper, we provide a brief overview.

The algorithm propagates *labels* from the source node to the sink node. A label represents a partial path and is associated with a node. It stores the accumulated cost and resource consumptions along that path. A label L_i associated with node i can be extended to a node j if and only if arc (i, j) exists. In that case, a new label L_j is created and associated with node j. The cost and resource consumptions of L_j are obtained by adding the cost and resource consumptions of arc (i, j) to those of L_i. Upon arrival at node j, resource values are increased, if necessary, to satisfy the soft lower bounds of the resource windows at node j. The resulting label is retained only if it respects the upper bounds of the resource windows at node j.

The nodes are processed according to a topological ordering of the network. Before extending labels at a node i, a dominance procedure is applied to discard dominated labels. A label L^1 is dominated by a label L^2 if both labels are associated with the same node and if the cost and all resource consumptions of L^2 are greater than or equal to those of L^1.

At termination, the sink label with the lowest cost corresponds to an optimal path. Sink labels with negative cost correspond to variables with negative reduced cost and can therefore be added to the restricted master problem (RMP).

4.1.4. Alternative network formulations

Note that the proposed network structure is not trivial and has several advantages. In particular, charging is modeled as to minimize the number of arcs in the network. A more trivial implementation would instead include charging decisions on trip-to-trip arcs. Several arcs would connect trip node i to trip node j, each corresponding to a different charging start time and duration. However, such a

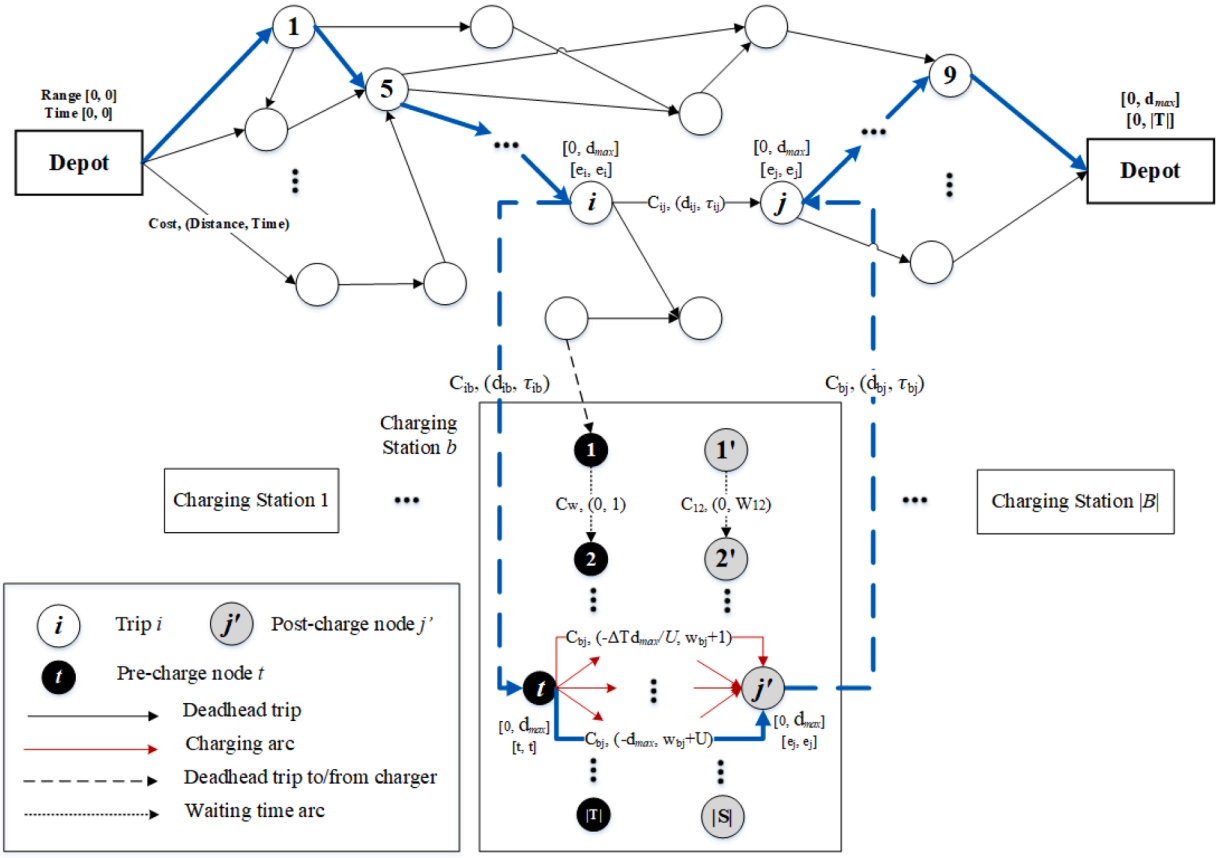


Fig. 2. Acyclic single-depot network for the pricing subproblem, showing pre-charge and post-charge nodes of a fast-charging station with a sample electric bus path with blue color.

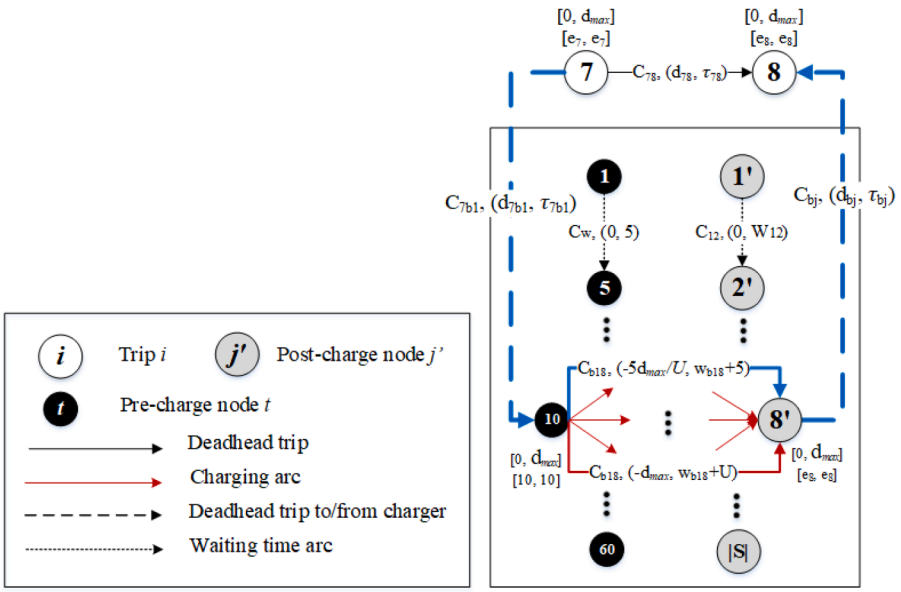


Fig. 3. An illustrative example of fast-charging station component of the acyclic network.

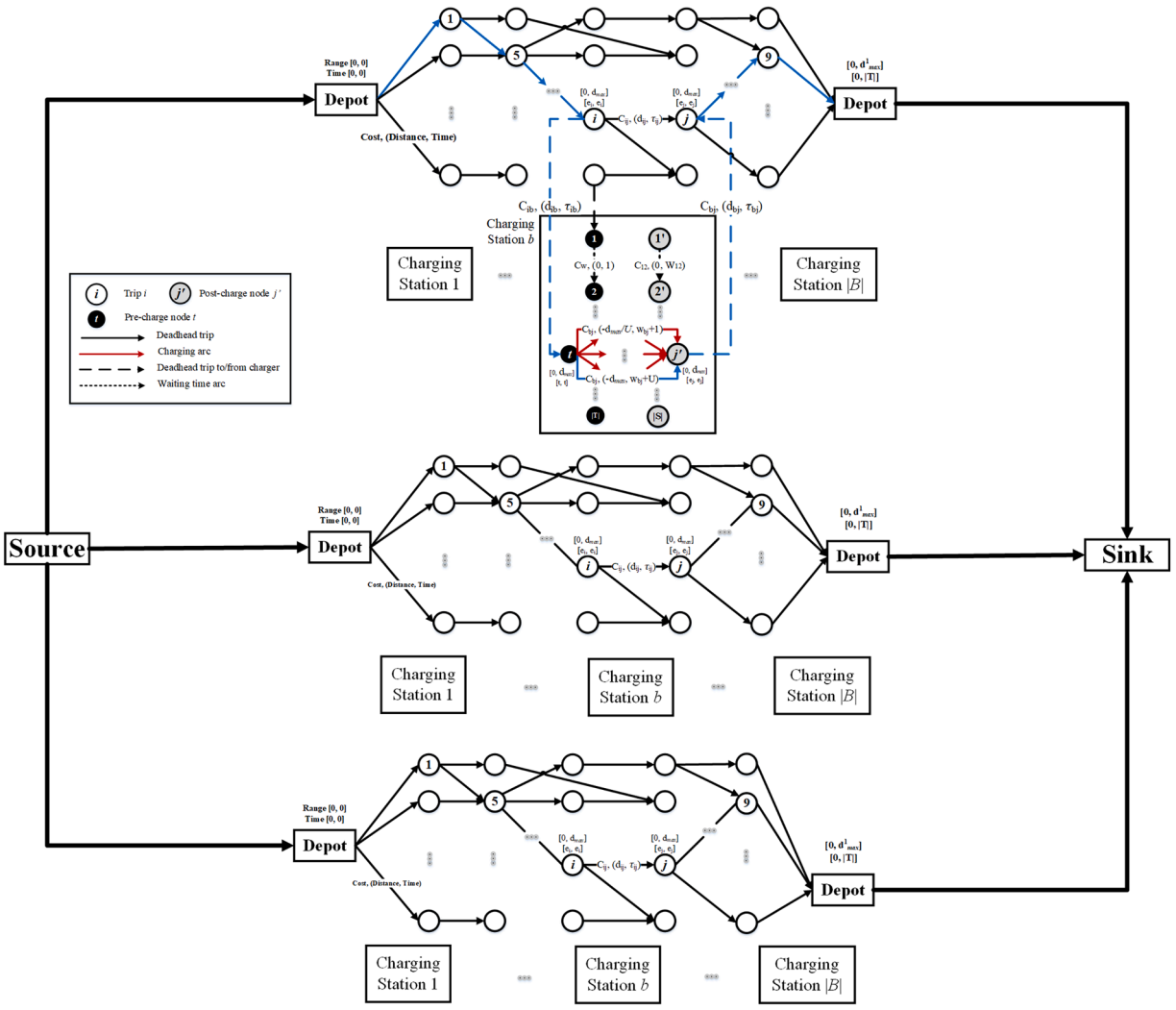


Fig. 4. Acyclic multi-depot network for the pricing subproblem, showing three depots.

network structure would yield $\mathcal{O}(|S|^2|B|\lceil U/\Delta T\rceil)$ arcs, whereas the proposed structure has $\mathcal{O}(|S||B|+|T||B|)$ arcs (Each trip has one arc going to and coming from each fast-charging station, and each fast-charging station has $\mathcal{O}(|T|)$ arcs connecting the pre-charge nodes and post-charge nodes among themselves). Since in practice the time required to solve the constrained shortest-path problem grows roughly quadratically with the number of arcs, this network reduction represents significant time savings.

In the proposed network structure, in addition to the $\mathcal{O}(|S|^2)$ trip-to-trip arcs, each trip node has exactly one outgoing arc to each fast-charging station, which connects the trip to the earliest feasible time-expanded node. This results in $O(|S||B|)$ arcs. For each fast-charging station, the time-expanded representation includes $|T|\cdot(U/\Delta T)$ charging arcs, as well as a linear number of waiting arcs. These waiting arcs allow vehicles to remain idle either at trip nodes or at charging stations. The total number of waiting arcs for each fast-charging station is $|T|+|S|-2$ and does not affect the asymptotic complexity.

Another alternative network structure would use a single node to represent each fast-charging station, with deadhead arcs between the trip nodes and the fast-charging station nodes. This formulation also has several drawbacks. Firstly, several arcs would be required between trip node $i\in S$ and fast-charging station $b\in B$, each one corresponding to a different charging schedule (duration and beginning time). Secondly, the network structure would no longer be acyclic, which would complicate the label-setting algorithm. Finally, resource extension and label dominance rules would have to be adapted. For instance, some labels associated with a given fast-charging station may not be comparable.

Finally, in the multi-depot setting, one could alternatively formulate one single-depot subproblem per depot, instead of a single multi-depot subproblem. This alternative formulation was not investigated. However, similar computational performance is expected. Indeed, the multi-depot subproblem decomposes into several independent networks, one per depot, so that the label-setting algorithm effectively solves a separate subproblem for each depot.

4.2. Branching

To achieve an integer solution, the branch-and-price algorithm embeds column generation in a branch-and-bound tree. Exact branch-and-price approaches can lead to prohibitive computational runtimes on large-scale instances. In those cases, heuristic strategies can be used to speed up the solution process while sacrificing little in terms of solution quality.

In this paper, we explore one exact and three heuristic branching strategies. As such, we use three branching strategies, such as bound and inter-task fixing (BITF), multiple column fixing (MCF), and the combination of MCF and BITF. The first one is an exact method that branches on static variables and inter-tasks. The second strategy is a column fixing heuristic, which sets one or more path variables to 1 at each node in a diving fashion. The third strategy mixes the exact method with column fixing. Also, we implement an acceleration strategy based on the solution improvement during CG process and its number of iterations threshold. The aim is to find a strategy that offers a good compromise between objective value and CPU time.

4.2.1. Exact branching (BITF)

Our exact branching method works by branching on static variables (i.e., the z variables determining fast-charging station locations) and inter-tasks (network arcs that link two tasks). Consider the linear relaxation of (1) to (5) at a given node. The BITF method prioritizes branching on static variables with fractional values, z, if there are any. The score for a static variable is calculated based on its fractional value at the time of evaluation. Specifically, let $\hat{z} = \mathrm{z} - \lfloor \mathrm{z} \rfloor$, and the score is given by: $2|\hat{z} - 0.5|$. The variable with the highest score is then selected for branching.

If all static variables are integer, the algorithm instead branches on an inter-task arc in the subproblem network. The arc with cumulative flow closest to 0.5 is selected. Suppose arc (i, j) is the one selected. Two child nodes are then created, in which this inter-task arc is either forbidden or imposed. The former decision is enforced by removing arc (i, j) from all networks, while the latter is enforced by removing all arcs leaving i and all arcs entering j, except arc (i, j). In both child nodes, all columns that contradict the decision are also removed (Desaulniers et al., 1997).

4.2.2. Multiple column fixing (MCF)

The column fixing algorithm fixes one of several path variables at each node without backtracking. To enforce path p on one branch, it is necessary to eliminate all paths that include the trips covered by p from the RMP, and to remove all arcs or nodes that represent those trips from the subproblem networks. The method selects which variables to fix based on their value in the previous node's fractional solution. It first fixes the variables with the highest fractional value. This selection method is used because columns with larger fractional values tend to have a smaller impact on the objective function when fixed at 1, compared to those with smaller values (Saddoune et al., 2013). This technique, known as the diving heuristic, follows a depth-first approach, continuing down a single branch of the tree until it finds a solution where all variables are integers. The primary benefit of this branching strategy is its ability to find feasible solutions quickly. However, the quality of these solutions may decrease, particularly in smaller instances, where fixing a variable can significantly affect the lower bound.

At each node, columns to be fixed are selected in decreasing order of their values, subject to the following rules. If no columns have a fractional value greater than or equal to a predefined threshold $\eta \in]0, 1[$, only the column with the largest value is selected. Otherwise, only columns with fractional values greater than or equal to η are considered for fixing. The choice of η plays a crucial role in the solution process: a large value of η significantly restricts the number of variables fixed at once, which can slow down the process, whereas a small value allows more variables, even those with relatively small fractional values, to be fixed simultaneously, which may sometimes lead to poor-quality solutions. In this research, if a column has a fractional value greater than or equal to 0.5, column fixing branching is applied. Otherwise, a combination of strong branching and column fixing is used (Quesnel et al., 2017).

To further control this process, the complementary value of a path variable y_p is defined as $1 - y_p$. The sum of the complementary values of the fixed columns must not exceed a maximum value $\psi \geq (1 - \eta)$, ensuring that the number of variables fixed near η remains manageable. Additionally, the total number of columns that can be fixed at a time is restricted to a maximum of μ.

4.2.3. Column fixing and exact branching combination

The third branching strategy, known as MCF-BITF, combines the MCF and BITF branching methods. In this heuristic approach, both strategies are applied simultaneously, and the one with the higher score is selected for branching. This process continues until all nodes are explored. The score for MCF is determined by the highest fractional value obtained. The score obtained from exact branching (BITF) is normalized and mapped within the range of $[b_l, b_u]$. Finally, the branching method with the highest score is selected.

4.2.4. Acceleration strategy

Column generation typically converges slowly as it approaches the optimal value, a phenomenon known as the tailing-off effect. At this stage, adding paths with negative reduced costs to the model often has little effect on the value of the RMP. To counter this issue, we choose to stop column generation prematurely if there is less than κ percent improvement in the RMP's optimal value over the last β iterations. The downside of this acceleration strategy (AC) is the loss of the optimality guarantee normally offered by CG. However, in our case, this has a negligible effect on the optimality gap. In our experiments, we consider using this acceleration strategy in combination with MCF-BITF branching strategy.

Table 2
Parameter settings for generating random instances.

Parameter	Description	Value	Reference
C_v	Annualized cost of a new bus	$ 35,333	Bloomberg New Energy Finance (2018)
V_b	Annual cost of establishing fast chargers	$ 15,333	Bloomberg New Energy Finance (2018)
C_d	Unit travel cost	0.06 $/km	Li (2014)
U	Charging time	10 min	Li (2014)
C_w	Cost of waiting time	0.01 $/min	
d_{max}	Maximum travel range of EBs	120 km	
T	Set of time periods	$\max_{i \in S}\{e_i\}$	

5. Results and analysis

We provide details on the experimental protocols used for our computational results in Section 5.1. We present comparative results of the proposed branching methods and acceleration strategy for instances of different sizes and a comparison of sequential and integrated approaches for solving the problem, in Section 5.2. Then, in Section 5.3, the Montreal case study is discussed to further assess the effectiveness of the proposed model. A brief sensitivity analysis across different types of EBs will be presented to evaluate the impacts of changing such parameters' values on the number of required EBs, fast-charging stations and total costs. In addition, the impact of timestep resolution on the objective value and the runtime will be discussed. Also, we conduct a sensitivity analysis of the main branch-and-price parameters in Section 5.4. Finally, Section 5.5 presents a comparison between the developed branch-and-price algorithm and the arc-based model.

5.1. Experimental protocol

For our computational experiments, most of the parameters used to generate instances of varying sizes are drawn from Li (2014) and Gkiotsalitis et al. (2023). The starting and ending points of the trips, the coordinates of the depots, and the coordinates of candidate locations for locating fast chargers are located in a 10 km to 10 km square on the Euclidean plane, with an even distribution. The travel time between any two points is proportional to the Euclidean distance between them. Based on Bloomberg New Energy Finance (2018), the capital cost of pantograph chargers and electric buses are $230,000 and $530,000, respectively. The lifetime of both EBs and pantograph chargers is 15 years. Thus, the annualized purchase cost of an EB and the annualized installation cost of fast chargers will be around $35,333 and $15,333, respectively. The actual charger installation cost may vary by location and is multiplied by a factor in the range [1, 1.25]. In this study, we assume $W = 1$, meaning that each station can support only one charging operation at a time. The model, notation, and constraints remain valid and directly applicable for any value of W. Also, the maximum travel range of EBs is assumed to be 120 km. This information and other required parameters are given in Table 2. It is important to note that the operational costs shown in Table 2 will be multiplied by the number of service days in a year. This will convert them into annual costs, allowing for a direct comparison with the yearly expenses of acquiring a new bus and installing a fast-charging station.

To solve the proposed model, we utilized the specialized Gencol library (see Desrosiers et al. (2024) for a comprehensive description and implementation details) version 4.5, which is specifically designed for the branch-and-price algorithm on a Linux computer with an Intel i7-8700 CPU clocked at 3.20 GHz without parallel computing. All RMPs are solved using the primal simplex algorithm with CPLEX 22.1.1. The time limit for solving large-scale instances is 48 h.

The branch-and-price parameters were manually tuned to achieve a good balance between solution quality and computational time. As shown by the sensitivity analysis in Section 5.4, the algorithm performance is only weakly affected by the choice of parameter values.

5.2. Results on synthetic instances

Each instance is encoded according to its number of trips, S, candidate spots to locate fast chargers, B, and the number of time periods, T. For example, S30-B5-T80 indicates that there are thirty trips, five candidate places to install fast chargers, and 80 time periods for each fast-charging station. The timestep for pre-charge nodes is set to 5 min, meaning that the start charging time must be in multiples of this interval. We solve instances of different sizes for two depots. Instances are categorized based on the number of trips: those with fewer than 100 trips are labelled as small, instances with more than 100 trips and less than 300 trips are categorized as medium-scale, and, finally, instances that involve more than 300 trips are classified as large-scale instances. They are important to this study because smaller instances generally result in larger integrality gaps.

5.2.1. Comparison of branching strategies

In this section, we analyze and compare the performance of the four proposed methods for addressing the FCLP-EBS problem. To evaluate the methods, several problem instances of varying scales were generated and solved using each approach. The results of these experiments are summarized in Table 3, which provide a detailed comparison of the effectiveness of the branching and acceleration strategies. The comparison focuses on two key metrics: CPU time and optimality gap. The best relaxation cost is represented by Z_{relax}, while the best feasible solution is denoted by Z_{best}. The integrality gap is calculated as $|Z_{\text{best}} - Z_{\text{relax}}|/Z_{\text{relax}}$. The gap reported in Table 3 reflects the range of the optimality gap based on the calculated integrality gap. Since the optimality gap is smaller than or equal to

Table 3
Comparison of different branching and acceleration strategies in terms of CPU time and optimality gap.

Instance	BITF		MCF		MCF-BITF		MCF-BITF-AC	
	Time (s)	Gap (%)	Time (s)	Gap (%)	Time (s)	Gap (%)	Time (s)	Gap (%)
S30-B5-T80	0.3	**0.000**	0.3	< 1.423	0.3	< 1.423	0.2	< 1.427
S30-B10-T80	2.1	**0.000**	2.3	< 1.832	1.2	< 1.701	0.5	< 1.708
S50-B5-T88	0.9	**0.000**	0.8	< 0.701	1.8	< 0.699	0.8	< 0.699
S50-B10-T88	36.1	**0.000**	10.3	< 2.401	7.3	< 2.296	5.9	< 2.351
S100-B5-T88	124.6	**0.000**	6.0	< 0.873	3.4	< 0.810	2.4	< 0.809
S100-B10-T88	77.8	**0.000**	16.4	< 0.007	16.2	< 0.007	5.3	< 0.003
S100-B20-T88	1864.0	**0.000**	26.2	< 1.094	943.2	< 0.920	188.3	< 0.922
S200-B10-T108	–	–	52.6	< 1.826	828.2	< 0.551	312.5	< **0.496**
S200-B20-T108	–	–	166.6	< 2.672	8931.0	< **0.648**	2296.6	< 0.678
S300-B10-T161	–	–	401.8	< 2.398	5706.0	< 0.723	2858.0	< **0.497**
S300-B20-T161	–	–	1547.9	< 0.846	–	–	74146.5	< **0.581**
S400-B10-T161	–	–	690.0	< 1.890	34171.0	< 0.518	6560.0	< **0.395**
S500-B10-T161	–	–	1451.0	< 0.716	59604.0	< **0.310**	17083.0	< 0.395
S500-B20-T161	–	–	4220.0	< **2.359**	–	–	–	–
S1000-B10-T229	–	–	36102.0	< **0.220**	–	–	–	–
Average		0.000		1.417		0.883		0.843

the integrality gap, we can conclude that the obtained solutions have an optimality gap at most equal to the integrality gap. Table 3 highlights the strengths and weaknesses of each method, offering insights into their computational efficiency and solution quality. By comparing these methods, we aim to identify the most practical and efficient approach for solving the FCLP-EBS problem across a range of scenarios.

Table 3 shows the computational results used to compare the performance of the four methods. For each instance, we report the computational time and the optimality gap range, which is calculated through the integrality gap between the best feasible solution and the relaxed solution. Since all methods produce identical linear relaxation solutions, these gaps allow us to compare the quality of solutions obtained by each heuristic. However, to ensure consistency, the integrality gap is calculated using the relaxed solution from the other methods. In Table 3, the branching strategy with the lowest gap in each row is highlighted in bold. Additionally, the last row of the table shows the average optimality gap for each branching strategy.

The BITF strategy consistently achieves a zero optimality gap in small-sized instances, demonstrating its effectiveness in finding optimal solutions for smaller problem sizes. However, its runtime becomes excessive as the problem size increases, since BITF only branches on fast-charging station variables and inter-task network arcs, rather than directly on columns or paths. In contrast, MCF has the lowest runtime but also the highest average gap (1.417%), making it less suitable for small and medium instances where solution precision is important. This is because MCF quickly finds feasible solutions, but fixing variables too early can significantly affect the lower bound, especially in smaller instances. MCF-BITF offers a balanced trade-off, with improved accuracy (average gap of 0.883%) and moderate runtime. As a hybrid of exact branching and MCF, it is more computationally intensive than MCF alone, especially in medium and large instances. The MCF-BITF-AC strategy applies an acceleration technique to MCF-BITF, achieving a similar level of accuracy (average gap 0.843%) while reducing runtime, making it the most effective choice for large-scale instances. These results suggest that BITF is ideal for small instances, MCF-BITF-AC is preferable for large-scale instances where runtime is a concern, and MCF is suitable for very large problems where fast, approximate solutions are acceptable.

The summary of the solving process of the proposed branch-and-price algorithm for different instances is shown in Table 4. It highlights the efficiency and computational performance of four strategies, detailing key metrics such as the number of Branch-and-Bound nodes (BB), Column Generation Iterations (CGI), and Columns Generated (CG).

The number of branch-and-bound nodes reflects the depth of the search process under each branching strategy. As shown in Table 4, the BITF strategy solves only small instances efficiently and it becomes infeasible as the problem size increases. In contrast, the MCF strategy solves more linear relaxation problems, suggesting broader but less focused search behavior. The MCF-BITF strategy significantly increases the number of solved relaxations, indicating deeper exploration that improves solution quality but leads to longer runtimes. Notably, MCF-BITF-AC records the highest BB in large instances, demonstrating its ability to thoroughly explore complex solution spaces while keeping runtime manageable, thanks to the adopted acceleration mechanism. Importantly, the number of BB rises sharply with an increase in candidate charger locations, indicating that the branching strategies are more sensitive to the number of candidate spots for locating chargers than to the total number of trips.

The number of columns generated is an indicator of the method's approach to refining the solution. MCF-BITF-AC and MCF-BITF tend to produce more columns compared to the MCF strategy as the instance size increases, suggesting that these strategies use iterative refinement intensively to converge on the optimal solution. The MCF-BITF-AC strategy generally performs well even with lower CGI and CG values on larger instances compared to the BITF and MCF-BITF methods, indicating its efficiency in handling large-scale instances. BITF tends to have higher values in small instances, especially in CG. The reason the number of CGI and CG in all branching strategies except MCF-BITF-AC are the same for the first instance, S30-B5-T80, is that the root node is already integer. Another point is that the effect of increasing the candidate spots for locating chargers is much more than increasing the number of trips in terms of CG.

Table 4
Solving process results. CGI: Column Generation Iterations, BB: Branch-and-Bound nodes, CG: Columns Generated.

Instance	BITF			MCF			MCF-BITF			MCF-BITF-AC		
	BB	CGI	CG	BB	CGI	CG	BB	CGI	CG	BB	CGI	CG
S30-B5-T80	7	184	586	7	184	586	7	184	586	15	110	492
S30-B10-T80	15	1386	2349	7	344	915	42	737	1489	42	386	1018
S50-B5-T88	9	369	1603	29	469	1272	9	369	1603	18	311	1421
S50-B10-T88	49	1958	4570	22	717	2075	83	2101	4518	65	1047	3658
S100-B5-T88	5	235	1959	90	896	2709	73	798	2902	66	539	2334
S100-B10-T88	23	1391	8418	66	1250	4948	85	2290	8791	78	834	4692
S100-B20-T88	185	30,878	191,296	30	2135	10,494	316	30,878	191,296	181	2287	24,404
S200-B10-T108	–	–	–	31	1830	14,150	207	3999	41,390	152	1876	19,453
S200-B20-T108	–	–	–	32	2879	23,395	350	77,286	839,240	734	11,835	161,163
S300-B10-T161	–	–	–	41	2692	22,062	114	20,960	227,037	748	33,388	196,083
S300-B20-T161	–	–	–	61	12,888	78,035	–	–	–	2527	506,937	1,309,127
S400-B10-T161	–	–	–	50	3823	36,720	183	47,249	757,314	356	7998	137,977
S500-B10-T161	–	–	–	43	5918	51,572	404	49,268	772,068	802	15,486	325,566
S500-B20-T161	–	–	–	59	8689	87,047	–	–	–	–	–	–
S1000-B10-T229	–	–	–	67	24,700	253,413	–	–	–	–	–	–

Table 5
Comparing the integrated with the sequential approach for EB scheduling and charging location planning.

Instance		Scheduling Cost	No. EB	Total Cost	Cost Deviation (%)
S50-B5-T88	Integrated	1,001,709	15	1,018,008	–
	Add	1,001,672	15	1,033,726	1.54
	Remove	1,725,778	26	1,757,832	72.67
	Change	1,200,288	18	1,216,043	19.45
S100-B5-T88	Integrated	1,769,232	26	1,803,274	–
	Add	1,766,721	26	1,817,062	0.76
	Remove	1,900,889	28	1,916,644	6.28
	Change	1,900,073	28	1,932,127	7.14
S70-B15-T88	Integrated	1,337,548	20	1,371,100	–
	Add	1,337,039	20	1,386,005	1.08
	Remove	1,469,999	21	1,486,574	8.42
	Change	1,403,437	21	1,435,426	4.96
S100-B20-T88	Integrated	1,742,675	26	1,775,744	–
	Add	1,337,039	20	1,386,005	1.08
	Remove	1,809,305	27	1,824,910	2.76
	Change	1,807,751	27	1,838,770	3.54

5.2.2. Sequential vs. integrated approach

We evaluate the benefits and potential improvements of the integrated approach compared to the sequential approach. The sequential approach solves two separate problems: first, identifying fast-charging station locations, and second, addressing the standard EB scheduling problem. In practice, this initial step is often performed manually by an experienced planner, relying on intuition rather than optimization. To simulate the decisions of a skilled but imperfect planner, we first solve the integrated model to determine the optimal fast-charging station locations. Next, we intentionally modify these locations in one of three ways: first, adding a fast-charging station, second, removing a fast-charging station, and finally, relocating a fast-charging station to a new selected location with the minimum installation cost. For each altered configuration, we resolve the EB scheduling problem using the proposed FCLP-EBS model, keeping the fast-charging station decisions fixed. For all modification types, we test all possibilities, but only present the results for the one which has the least impact on the overall solution cost. This simulates the decision process of a nearly-perfect planner. This process allows us to assess the robustness and efficiency of the integrated model compared to sequential decision-making. This experiment was conducted on four instances of varying scales, and the resulting percentage cost deviations for each instance and scenario are summarized in Table 5.

The results of Table 5 illustrate the substantial impact of fast-charging station location planning on operational costs across various instances. Removing fast-charging stations often leads to notable cost increases; for example, the first instance experienced a nearly 73% cost rise, highlighting the critical role of infrastructure in maintaining operational efficiency. In contrast, planning too many fast-charging stations has a limited impact. This is expected since, in the worst case, the fast-charging station is never utilized and the only added cost of the solution is the establishing cost, V_b. Modifying fast-charging station locations resulted in mixed outcomes, with some instances showing moderate cost increases. This variability emphasizes the complexity of optimal placement, indicating that even minor adjustments can have significant financial consequences. Careful planning and analysis are therefore essential to optimize electric bus operations effectively.

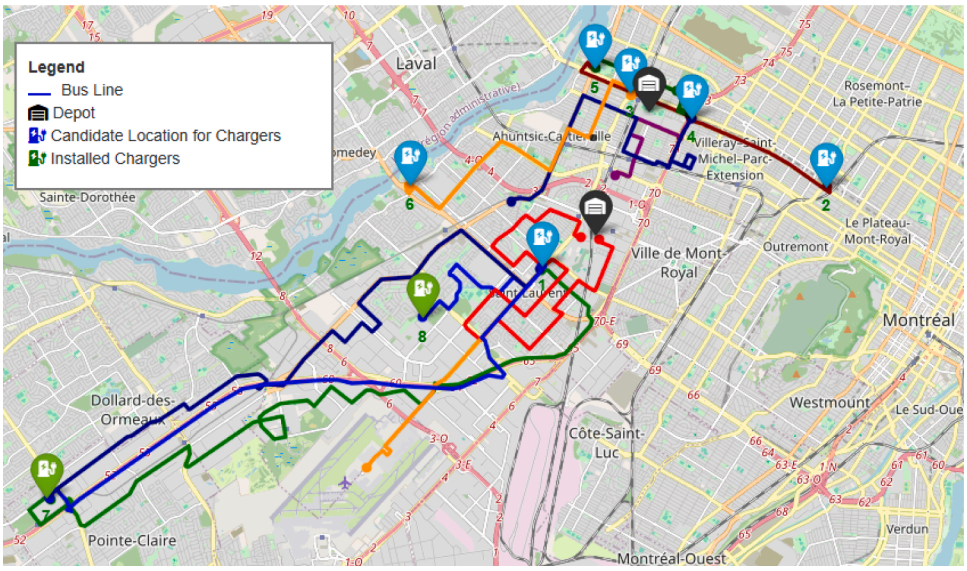


Fig. 5. Geographical distribution of the selected bus lines departing from two depots and candidate spots for installing fast-charging stations.

Table 6
Information about transit routes and trips.

Bus Line	Travel Time (min)	Route Length (km)	Departure Frequency (min)
70	28	9.4	20
72	51	17	15
19	16	5.4	20
128	63	21	30
174	21	7	25
31	26	8.7	30
135	33	11	30
180	31	10.4	20
56	8	2.7	15
146	26	8.7	30
215	49	16.4	25
216	31	10.4	27

5.3. Case study

This section evaluates the performance of the proposed model and algorithm through a real-life case study in Montreal, Canada. We focus on two major bus depots within the city, Garage Stinson and Garage Legendre, and twelve bus lines are assigned to these depots for potential electrification. Using data from the General Transit Feed Specification (GTFS) of Montreal, each bus line's operational characteristics, such as departure frequency, route length, and travel time, are presented in Table 6. Fig. 5 illustrates the layout of the transit network under study, which features the bus routes and eight candidate locations for fast-charging stations. Electric buses on these routes are initially stationed at the two depots, marked by black icons on the map. EBs can serve trips on different routes if they remain cost-efficient.

The total number of trips considered across all bus lines is 507. Electric buses are allowed to recharge at any point during the day, from the beginning to the end of their service hours. The cost of purchasing and installing a fast-charging station is estimated at 500,000 CAD, although this cost may vary depending on the station's location. The assumed lifespan of each charger is 12 years. In terms of operational expenses, buses incur a cost of 0.17 CAD per kilometre travelled. The charging cost with fast chargers is set at 15 CAD per hour during peak hours and 30 CAD per hour during off-peak hours. Peak hours are defined as 7:00 AM to 11:00 AM and 5:00 PM to 7:00 PM.

Table 7 illustrates the characteristics of four real-world electric bus types according to Proterra Electric Vehicle Technology Manufacturer (2016b) and Gallo et al. (2014). The differences in electric bus types are based on their purchasing cost, maximum travel range, and charging time. Note that the lifetime of all EBs is assumed to be 12 years. We conduct individual model runs for each electric bus type for the Montreal case study and present the corresponding results. This analysis can assist transit authorities of Montreal in determining the most suitable electric bus type for their fleet, in terms of cost-effectiveness.

The results of the case study obtained using the MCF-BITF-AC method for different types of electric buses are presented in Table 8. It highlights the optimum number of EBs required to fulfill the given trips and the index of the selected locations for installing chargers.

Table 7
Characteristics of four well-known electric bus types (Proterra Electric Vehicle Technology Manufacturer, 2016a; Gallo et al., 2014).

Electric Bus	Price (CAD)	Annual EB Usage Cost (CAD)	Maximum Travel Range (km)	Charging Time (min)
New Flyer	852,500	71,041	90	6
Proterra Catalyst	958,500	79,875	116	10
Proterra Catalyst +	1,035,000	86,250	140	13
Proterra Catalyst XR	1,110,000	92,500	263	13

Table 8
Comparison of results for different EB types for the case study.

Electric Bus	Charger Location(s)	No. EB	CAPEX (CAD)	Annual Bus and Charger Cost (CAD)	OPEX (CAD)	Obj (CAD)
New Flyer	7,8	29	25,722,500	2,143,542	1,743,032	3,886,574
Proterra Catalyst	6,8	28	27,838,000	2,319,833	1,697,705	4,017,538
Proterra Catalyst +	7	27	28,445,000	2,370,417	1,637,034	4,007,451
Proterra Catalyst XR	–	27	29,970,000	2,497,500	1,613,328	4,110,828

The table also reports the total capital expenditures (CAPEX) for purchasing EBs and chargers, as well as the annualized cost of these investments. The operational expenditures (OPEX) column indicates the total travelling cost, waiting time cost, and cost of charging. The last column presents the objective value. The objective value includes the total operational costs, electricity costs, and the annual cost of purchasing EBs and installing fast chargers. The aim is to find the EB type with the lowest total costs among the given electric bus types. By conducting these comparisons, we can identify the most cost-effective electric bus type for the case study. The results of this analysis are represented in Table 8.

According to Table 8, the New Flyer bus type emerges as the most cost-efficient option, achieving the lowest objective cost among all alternatives. Although its shorter travel range increases the need for more frequent recharging, which requires a larger fleet and additional chargers to maintain service levels, its lower procurement cost significantly reduces the overall capital expenditure. Additionally, the New Flyer benefits from a shorter charging time, allowing EBs to recharge more quickly and return to service sooner. This added flexibility improves scheduling efficiency and increases the number of trips that each bus can cover, further strengthening its overall cost-effectiveness. In contrast, the Proterra Catalyst XR, despite offering the longest travel range, results in the highest objective cost. This is primarily driven by its substantially higher purchase price, which increases both CAPEX and the annualized bus and charger costs. While the extended range reduces the operational costs and the number of recharging trips and potentially the need for charging infrastructure, the cost savings do not offset the high investment required. The Proterra Catalyst and Proterra Catalyst + fall between these two extremes. They offer a balance between upfront investment and operational efficiency, yielding moderate objective values. Both require fewer chargers than the New Flyer, but still carry higher CAPEX due to their bus prices.

Another noteworthy observation is the variation in charger locations across EB types. For instance, the optimal charger placement for New Flyer is at spots 7 and 8, while Proterra Catalyst places them at 6 and 8, and Catalyst + requires only a single location at location 7. In contrast, when deploying Proterra Catalyst XR buses, no fast chargers are needed on the network, as these buses have sufficient range to be fully charged overnight at the depots, eliminating the need for mid-day charging infrastructure. This variation underscores the importance of considering bus characteristics, such as range and charging duration, when planning charging infrastructure. Charger location planning must be tightly integrated with fleet selection to ensure efficient operation across the entire network. Overall, these findings highlight the trade-offs between initial investment, operational flexibility, and infrastructure requirements. Longer-range buses reduce recharging frequency and infrastructure needs but come at a higher upfront cost. Shorter-range buses may require more logistical coordination but offer lower total costs when supported by efficient charging and scheduling strategies. This analysis emphasizes the critical role of travel range and charging efficiency in optimizing the deployment of electric buses in public transportation networks.

Fig. 6 shows the breakdown of deadhead time, waiting time, and charging time for each type of electric bus. The figure highlights how operational time varies across different EB types. It is important to note that the total in-service time for all buses is 32,550 min.

The analysis of deadhead, waiting, and charging times reveals important differences among the electric bus types. The New Flyer has the highest charging time and a relatively high deadhead time, likely due to its shorter range requiring more frequent trips to fast-charging stations. However, its lower procurement cost and faster individual charging sessions can improve scheduling flexibility. In contrast, the Proterra Catalyst XR shows the lowest deadhead and waiting times, indicating higher operational efficiency due to its longer travel range and reduced need for frequent charging. With no charging time at all, it is particularly well-suited for minimizing downtime. Proterra Catalyst + experiences the highest waiting time, suggesting less efficient vehicle utilization. Overall, the results highlight the trade-offs between travel range, charging needs, and total operational time, which should be carefully balanced when selecting an EB type.

Fig. 7 complements Tables 7 and 8 by providing a clear visual comparison of electric bus alternatives across vehicle characteristics and model outcomes. The heatmap uses normalized scores, where lighter cells indicate better performance, making trade-offs easy to spot. For example, the New Flyer performs well in purchase price and total annual cost, but it requires more buses and chargers. In contrast, the Proterra Catalyst XR offers the best travel range and reduces charging needs, but it leads to higher capital costs. The intermediate Proterra options show more balanced performance and achieve total costs close to the best solution, consistent

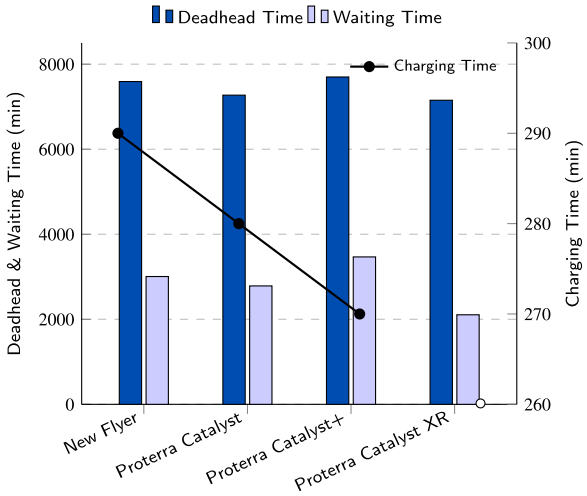


Fig. 6. Comparison of deadhead time, waiting time, and charging time for different EB types.

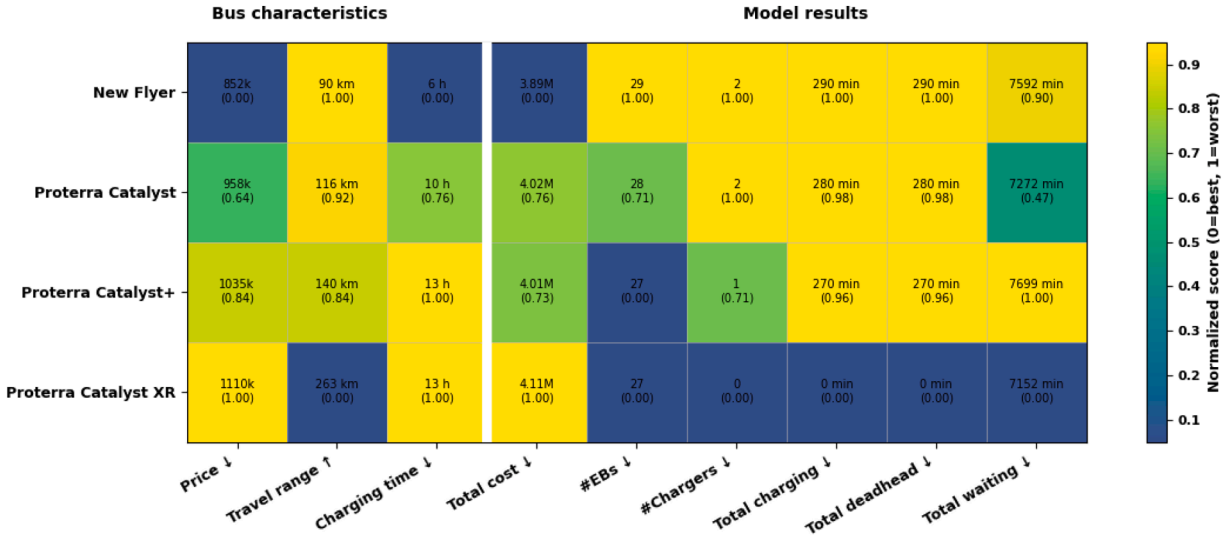


Fig. 7. Heatmap comparing bus alternatives using normalized scores across bus characteristics and model results.

with Table 8. Overall, the heatmap provides an intuitive summary of the sensitivity analysis and is recommended alongside tables to communicate key findings more effectively and support transparent decision-making.

This experiment was conducted in a specific region of Montreal, considering a fixed number of bus lines and candidate locations for installing chargers. It is important to highlight that the results may differ if the scale of the problem, such as the number of bus lines or candidate locations, changes. Therefore, it is essential to first determine the scale of the bus network, consider the number of given trips, and identify candidate locations for fast chargers. Once these factors are established, transit authorities can then assess which electric bus type best suits their fleet. By utilizing the proposed model and conducting a thorough analysis, transit authorities gain valuable insights to make well-informed decisions before purchasing or expanding their electric bus fleet. This approach empowers them to select the most suitable electric bus type that aligns with their specific requirements, resulting in a more efficient and cost-effective public transportation system.

5.3.1. Effect of timestep resolution on total cost and CPU time

The pre-charge nodes in our model use a default timestep of 5 min. To evaluate how changing this assumption affects both the solution quality and computational performance, we conduct an experiment using the New Flyer electric bus, which was identified in Table 8 as the most cost-effective option. We solve the problem using the MCF-BITF-AC branching strategy for two different timestep values: 5 min and 2 min. The goal is to understand the impact of a smaller timestep on the objective value and CPU time, and to determine whether the added computational effort is justified. Table 1 reports the total costs and runtime for both timestep settings.

Table 9
Comparison of solution quality and runtime for timesteps 2 and 5.

EB Type	Timestep = 2		Timestep = 5		Change in Total Costs (%)	Change in Runtime (%)
	Objective Value (CAD)	Time (s)	Objective Value (CAD)	Time (s)		
New Flyer	3,847,443	10,948	3,886,574	2321	1.00	–371.96

By comparing these results, we can assess whether the marginal gains in solution accuracy are worth the additional computation time, providing valuable insight into the trade-off between model precision and efficiency.

The results of Table 9 indicate that reducing the timestep from 5 min to 2 min leads to a slight improvement in solution quality, shown by a 1.00% decrease in the total costs. This gain reflects the finer modelling precision achieved with shorter timesteps, which allows more accurate scheduling of EBs and charging times. However, this modest improvement comes at a significant computational cost. The runtime increases sharply, from 2321 seconds at the 5-min timestep to 10,948 seconds at the 2-min timestep, representing a 372% increase in CPU time. This substantial rise in CPU time illustrates the higher complexity and size of the time-expanded network when a finer resolution is used. Overall, these findings illustrate a clear trade-off between accuracy and efficiency. Although the 2-min timestep produces a slightly better solution, the significant rise in computational time may not be justified, especially in large-scale or time-sensitive scenarios. In practical terms, the default 5-min timestep strikes a better balance, offering near-optimal results while keeping computational effort at a manageable level.

5.4. Sensitivity analysis

We conduct a sensitivity analysis on the branching-strategy parameters for the MCF-BITF-AC method, to identify the combinations that minimize both runtime and integrality gap. This analysis is conducted on three instances of different sizes (S100-B5-T88, S100-B10-T88, and S200-B10-T108), allowing us to observe how parameter choices behave at varying scales and mitigate bias from tuning on a single instance size. Table 10 reports, for each instance, the solution time and final integrality gap obtained over a grid of parameter values $(\eta, \psi, \mu, b_l, b_u, \kappa, \beta)$ for the branching heuristics and acceleration strategy. The first line corresponds to the selected parameter values.

According to Table 10, the average runtime and integrality gap vary only slightly across the tested parameter settings, suggesting that the algorithm is not highly sensitive to these choices. Nevertheless, for our computational experiments and the case study, we select the parameter configuration that offers the best trade-off between runtime and integrality gap. As shown in Table 10, moderate branching and acceleration settings (the first parameter setting) consistently yield small integrality gaps with relatively low and stable runtimes. Based on these results, we fix $\eta = 0.5$, $\psi = 0.7$, $\mu = 5$, $b_l = 0.8$, $b_u = 1$ for the branching strategy and $\kappa = 0.2\%$, $\beta = 15$ for the acceleration strategy.

5.5. Results comparison: PBM vs. ABM

We propose a mixed-integer linear programming (MILP) formulation based on an arc-based model as another method to tackle the FCLP-EBS problem. The details of the arc-based model are provided in Appendix A. The arc-based model presented in Appendix A assumes a fixed charging duration equal to the full-charge time U. Accordingly, the computational comparison between the ABM and the PBM is carried out under this fixed charging assumption. This simplification is introduced solely to keep the formulation computationally tractable. Allowing partial charging in an arc-based framework would substantially increase the size of the model and the solution time. In fact, even with fixed charging duration, the arc-based model already exhibits very long runtimes and becomes impractical for medium and large instances. For this reason, we do not extend the arc-based formulation to include partial charging. The PBM is trivially adapted by removing all charging arcs except for those corresponding to full charging. The fixed-duration version is sufficient to show that the arc-based approach is far less efficient than the proposed path-based method. By contrast, our branch-and-price solution naturally supports partial charging through arcs of varying lengths while maintaining strong computational performance, and would therefore outperform an arc-based model with partial charging even more clearly.

We solve the arc-based model with the CPLEX solver and compare its results in terms of runtime with the results of the developed path-based model solved using the B&P method. We solve instances of varying sizes for two depots. For each instance, we provide information on the number of trips, candidate spots for fast charger installations, and the number of pre-charge nodes, representing the available time for charging using pantograph chargers. A total of 21 instances are generated, and their results are presented in Table 11. The second and third columns show the total number of installed chargers and the total number of required EBs, respectively. The next two columns present the objective value and execution time of CPLEX and the exact branching B&P method, respectively.

A comparison of the results reveals that the branch-and-price algorithm outperforms the arc-based model solved with CPLEX in terms of computational efficiency. In every case, the computational time for the branch-and-price method is shorter than the execution time for the CPLEX solver. The difficulty of solving the instances is greatly influenced by changes in their parameters. For instance, when three additional locations for fast chargers were added to the S40-B5-T404 case, resulting in the S40-B8-T404 instance, the CPU processing time approximately doubled, indicating an increase in complexity. In a more extreme case, adding ten additional trips to the S50-B3-T444 instance made it so challenging that the CPLEX solver could not find the optimal solution within a day. It is important to note that the optimality gap was set to zero for all instances. Also, according to Li (2014), the cost of waiting time

Table 10
Sensitivity analysis on branching strategies parameters.

Parameter values							Instance S100-B5-T88		Instance S100-B10-T88		Instance S200-B10-T108		Average values	
η	ψ	μ	b_l	b_u	κ (%)	β	Time (s)	Gap (%)	Time (s)	Gap (%)	Time (s)	Gap (%)	$\hat{T}$ (s)	$\hat{\text{Gap}}$ (%)
0.5	0.7	5	0.8	1.0	0.2	15	2.4	0.809	5.3	0.003	312.5	0.496	106.7	0.436
0.3	0.7	3	0.6	1.0	0.3	20	2.6	0.810	6.7	0.000	286.3	0.557	98.5	0.456
0.3	0.7	3	0.6	1.0	0.1	10	2.6	0.813	12.4	0.010	407.7	0.599	140.9	0.474
0.3	0.7	3	0.5	0.7	0.3	20	2.6	0.825	6.3	0.012	346.2	0.589	118.4	0.475
0.3	0.7	3	0.5	0.7	0.1	10	2.4	0.822	6.2	0.023	360.5	0.583	123.0	0.476
0.3	0.7	7	0.6	1.0	0.3	20	2.6	0.809	6.8	0.000	307.4	0.557	105.6	0.455
0.3	0.7	7	0.6	1.0	0.1	10	2.3	0.804	12.0	0.010	373.0	0.552	129.1	0.455
0.3	0.7	7	0.5	0.7	0.3	20	2.8	0.822	8.2	0.009	297.1	0.585	102.7	0.472
0.3	0.7	7	0.5	0.7	0.1	10	3.1	0.826	7.5	0.003	314.9	0.568	108.5	0.466
0.3	1.0	3	0.6	1.0	0.3	20	3.0	0.809	6.8	0.000	291.8	0.562	100.5	0.457
0.3	1.0	3	0.6	1.0	0.1	10	2.7	0.809	13.0	0.035	274.4	0.546	96.7	0.463
0.3	1.0	3	0.5	0.7	0.3	20	2.6	0.820	7.2	0.013	333.8	0.551	114.5	0.461
0.3	1.0	3	0.5	0.7	0.1	10	2.6	0.818	6.2	0.006	328.5	0.601	112.4	0.475
0.3	1.0	7	0.6	1.0	0.3	20	3.3	0.806	6.8	0.000	345.9	0.550	118.7	0.452
0.3	1.0	7	0.6	1.0	0.1	10	3.0	0.803	12.4	0.013	257.9	0.555	91.1	0.457
0.3	1.0	7	0.5	0.7	0.3	20	3.6	0.819	11.3	0.007	376.4	0.552	130.4	0.459
0.3	1.0	7	0.5	0.7	0.1	10	4.0	0.819	10.7	0.003	246.0	0.568	86.9	0.463
0.7	0.7	3	0.6	1.0	0.3	20	2.4	0.809	6.8	0.000	270.3	0.557	93.2	0.455
0.7	0.7	3	0.6	1.0	0.1	10	2.4	0.814	11.2	0.040	245.6	0.554	86.4	0.469
0.7	0.7	3	0.5	0.7	0.3	20	2.8	0.824	6.6	0.028	316.1	0.571	108.5	0.474
0.7	0.7	3	0.5	0.7	0.1	10	2.5	0.824	5.5	0.016	258.2	0.584	88.7	0.475
0.7	0.7	7	0.6	1.0	0.3	20	3.0	0.813	6.8	0.000	291.2	0.566	100.3	0.460
0.7	0.7	7	0.6	1.0	0.1	10	2.3	0.803	11.8	0.005	355.6	0.583	123.2	0.464
0.7	0.7	7	0.5	0.7	0.3	20	3.6	0.828	7.8	0.012	344.8	0.569	118.7	0.470
0.7	0.7	7	0.5	0.7	0.1	10	2.9	0.838	7.1	0.003	253.5	0.586	87.8	0.476
0.7	1.0	3	0.6	1.0	0.3	20	2.6	0.809	6.9	0.000	303.8	0.549	104.4	0.453
0.7	1.0	3	0.6	1.0	0.1	10	2.4	0.814	11.4	0.038	307.9	0.550	107.2	0.467
0.7	1.0	3	0.5	0.7	0.3	20	2.8	0.824	6.6	0.028	358.6	0.595	122.7	0.482
0.7	1.0	3	0.5	0.7	0.1	10	2.5	0.824	5.5	0.016	262.2	0.591	90.1	0.477
0.7	1.0	7	0.6	1.0	0.3	20	3.0	0.813	6.8	0.000	297.4	0.564	102.4	0.459
0.7	1.0	7	0.6	1.0	0.1	10	2.4	0.808	15.7	0.038	245.0	0.561	87.7	0.469
0.7	1.0	7	0.5	0.7	0.3	20	3.8	0.827	7.5	0.015	364.5	0.564	125.3	0.469
0.7	1.0	7	0.5	0.7	0.1	10	3.2	0.824	7.0	0.003	261.0	0.588	90.4	0.472

Table 11
Comparison of the results of CPLEX and branch-and-price method.

Instance	No. Chargers	No. EB	Objective Value	Time (s)	
				CPLEX	B&P
S10-B3-T250	1	4	173445.1	1.5	0.1
S10-B5-T250	1	4	172568.3	2.3	0.2
S10-B8-T250	1	4	172699.7	4.3	0.7
S20-B3-T300	1	7	297467.8	84.8	1.7
S20-B5-T300	1	6	263147.5	32.5	1.2
S20-B8-T300	1	6	263059.9	65.4	4.1
S30-B3-T404	2	8	368904.5	117.5	2.8
S30-B5-T404	2	8	366700.4	205.4	16.4
S30-B8-T404	1	8	353743.2	255.9	76.5
S40-B3-T404	3	10	472012.9	91.8	9.3
S40-B5-T404	3	10	468976.1	193.7	31.7
S40-B8-T404	3	10	468740.8	393.9	152.8
S50-B3-T444	1	14	596099.5	604.3	53.3
S50-B5-T444	1	14	592956.9	615.2	124.2
S50-B8-T444	2	12	535677.5	1256.8	553.0
S60-B3-T444	2	15	659801.9	###	394.9
S60-B5-T444	2	15	656897.0	###	781.8
S60-B8-T444	2	15	654719.7	###	5043.5
S70-B3-T444	3	16	719046.7	###	564.2
S70-B5-T444	4	16	737987.1	###	12194.4
S70-B8-T444	3	16	721388.7	###	37105.6

is 0.1$/min , and the unit travel cost of EBs is 6.6 cent/km. Also, the maximum travel range of EBs is assumed to be 70 km. Other parameters are detailed in Table 2.

6. Conclusion

By rapidly replacing conventional buses with electric ones, new challenges to the operation planning of EBs will emerge. Given the limited travel range of EBs and their long charging durations, public transit operation planning necessitates rigorous reassessment. Electric bus scheduling is challenging because EBs have operating characteristics that differ from those of conventional buses. Additionally, location planning of fast-charging stations and scheduling the charging events further complicates decision-making. To address these complexities, this study introduced a path-based model through an integer-linear programming formulation and solved it with a branch-and-price algorithm. We evaluated the performance of the proposed branch-and-price algorithm through four branching strategies using instances randomly generated in a practical setting. Extensive computational experiments were conducted and identified the most effective branching techniques in terms of run time and optimality gap. Additionally, a real case study in Montreal, Canada was conducted to demonstrate the real-world applicability of the model. We solved the case study problem with real existing types of electric buses to help the transit authority of Montreal select the most suitable EB type with specific parameters for fleet electrification. This analysis highlights the model's potential to support transit agencies in identifying the EB type that best fits their transit system, ultimately reducing operational costs over the long term.

The proposed study has the potential for several extensions. One key area for improvement is to incorporate uncertainty into the model for EBs travel time and energy consumption. Travel time for EBs is influenced by various factors such as passenger volume, driving behavior, traffic congestion, road conditions, weather, and more. By considering stochastic travel time, the model can become more robust and better suited to real-world situations. Similarly, energy consumption for electric buses also varies depending on factors like passenger load, battery life, and road conditions. Future research could consider these factors to improve the accuracy of the model. Another area of interest to explore is the inclusion of different vehicle types in the proposed model. Currently, the model assumes that all the EBs in the fleet are homogeneous, while many cities have public bus transportation systems with heterogeneous fleets. Additionally, the model can be adapted to account for different charging technologies with varying establishment costs, charging power, and charging times.

To enhance the public transportation operation planning for electric buses, several additional elements could be added to the proposed model in future studies. For example, the second stage of public transportation planning, which is timetable development could be included as a decision variable, like the first and third stages, which are network design and bus scheduling, respectively. This means that departure times and trip frequencies could be optimized rather than being fixed parameters. Furthermore, since pantograph chargers have high electricity loads, it is important to consider the impacts of locating such facilities in neighborhoods. Future works could investigate the power loss incurred in connecting such infrastructure to the grid and its effects on energy sources.

Declaration of generative AI and AI-assisted technologies in the writing process

Generative AI was not used in a substantial manner for writing the manuscript.

CRediT authorship contribution statement

Kayhan Alamatsaz: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Conceptualization; **Frédéric Quesnel:** Writing – review & editing, Supervision, Methodology, Formal analysis, Conceptualization; **Ursula Eicker:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

Appendix A.

A.1. Arc-based model (ABM)

This section presents an arc-based mathematical formulation for the fast-charging location planning and electric bus scheduling (FCLP–EBS) problem. We model the capacity of a fast-charging station, denoted by W, using pre-charge nodes obtained by discretizing the range of possible charging start times. To ensure temporal feasibility of charging events, we define $T' = \{1, \dots, ((\max_T - U)/\Delta T) + 1\}$, which represents the set of feasible charging start periods. This definition ensures that any charging event, whether partial or full, can be completed within the discretized time horizon, given that the minimum charging duration is ΔT. Although, the arc-based model assumes a fixed charging duration of U. The set of pre-charge nodes associated with a candidate location $b \in B$ for installing a fast-charging station is denoted by T_b.

The set of arcs A is divided into five categories. There are depot-to-trip arcs between O and $j \in S$. There are trip-to-trip arcs between $j \in S$ and $i \in S$ if $e_j + \tau_{ji} \le a_i$. There are trip-to-charger arcs from $j \in S$ to pre-charge node t for candidate spot $b \in B$ if $e_i + \tau_{ib} \le max_T - U$, and charger-to-trip arcs from pre-charge node t for candidate spot $b \in B$ to $j \in S$ if $t + \tau_{bj} \le a_j$. Finally, there are trip-to-depot arcs between $j \in S$ and D. A' is the set of all possible arcs between trips and fast-charging stations and vice versa. The cost of arc ji is defined as: (1) $c_{ji} = C_d d_{ji} + C_v$ if $j = O$; (2) $c_{ji} = C_d d_{ji} + C_w w_{ji} + C_t$ if $i \in B$; and (3) $c_{ji} = C_d d_{ji} + C_w w_{ji}$ for other arcs.

The decision variables for the problem are as follows. Let z_b be a binary decision variable with $z_b = 1$ if a fast charger is located at location $b \in B$, and $z_b = 0$ otherwise. Let x^o_{ji} be a binary decision variable, with $x^o_{ji} = 1$ if an electric bus is assigned to node i immediately after trip j starting at depot $o \in O$, and $x^o_{ji} = 0$ otherwise. Similarly, f^{ot}_{jb} is a binary decision variable with $f^{ot}_{jb} = 1$ if a vehicle starting at depot $o \in O$ charges at the fast charger placed at location b directly after node j at time t, otherwise $f^{ot}_{jb} = 0$. Lastly, g^o_i is a continuous decision variable, representing the accumulated distance travelled from depot $o \in O$ to the ending point of trip $i \in S$ since the latest battery charging event.

The arc-based model can be formulated as a mixed-integer programming model as follows:

$$\min \left(\sum_{o \in O} \sum_{(j,i) \in A \setminus A'} C^o_{ji} x^o_{ji} + \sum_{o \in O} \sum_{(j,t) \in A'} C^o_{jt} x^o_{jt} + \sum_{b \in B} V_b z_b \right) \tag{A.1}$$

s.t.

$$\sum_{o \in O} \sum_{\substack{j \\ (j,i) \in A}} x^o_{ji} = 1 \quad \forall i \in S \tag{A.2}$$

$$\sum_{\substack{i \\ (j,i) \in A}} x^o_{ji} - \sum_{\substack{i \\ (i,j) \in A}} x^o_{ij} = 0 \quad \forall j \in S, \forall o \in O \tag{A.3}$$

$$\sum_{\substack{j \\ (j,t \in T_b) \in A'}} x^o_{jt} = \sum_{\substack{j \\ (j,t \in T_b) \in A'}} x^o_{tj} \quad \forall b \in B, \forall o \in O \tag{A.4}$$

$$g^o_i = \sum_{\substack{j \\ (j,i) \in A \setminus A'}} (g^o_j + d_{ji}) x^o_{ji} \quad \forall i \in S, \forall o \in O \tag{A.5}$$

$$g^o_t \le (1 - x^o_{jt}) d_{max} \quad \forall (j,t) \in A', \forall o \in O \tag{A.6}$$

$$g^o_i \le d_{max} \quad \forall i \in S \cup T, \forall o \in O \tag{A.7}$$

$$\sum_{o \in O} \sum_{\substack{i \\ (i,t \in T_b) \in A'}} x^o_{it} + \sum_{o \in O} \sum_{\substack{i \\ (i,t \in T_b) \in A'}} \sum_{s=1}^{U} x^o_{i,t+s} \le W \quad \forall b \in B, \forall t \in T' \tag{A.8}$$

$$\sum_{\substack{i \\ (i,t \in T_b) \in A'}} x^o_{it} \le z_b \quad \forall b \in B, \forall o \in O \tag{A.9}$$

$$x^o_{ji} \in \{0,1\} \quad \forall (j,i) \in A, \forall o \in O \tag{A.10}$$

$$z_b \in \{0,1\} \quad \forall b \in B \tag{A.11}$$

$$g_j^o \geq 0 \quad \forall j \in S, \forall o \in O \tag{A.12}$$

The objective function (A.1) includes three terms. The first term calculates the total annual travelling and waiting costs between trips plus the annual purchase cost of each electric bus, the second term determines the annual charging cost and travelling cost between trips and fast-charging stations, and the last term indicates the total annual establishment cost of fast chargers. Constraints (A.2) and (A.3) represent the covering and flow conservation of trips. Constraint (A.4) ensures that the inbound and outbound trips to/from a fast-charging stations should be equal. Constraint (A.5) determines the accumulative distance travelled since the latest battery charging event. This particular constraint introduces nonlinearity to the mathematical model. A linear version of our model can be found in Section A.2. Constraint (A.6) ensures that when an EB recharges at a station, its total travelled distance resets to zero. Constraint (A.7) forces that the total travelled distance of each EB cannot exceed its maximum travel range. Constraint (A.8) ensures that the station capacity constraint must be satisfied. Constraints (A.9) ensure that EBs can be charged only at the locations where a charger has been installed. Constraints (A.10) to (A.12) determine the domain of variables.

A.2. Linearization

The arc-based formulation in Section A.1 is nonlinear due to the multiplication of the binary variable, x_{ji}^o to the continuous variable g_j^o. We convert such formulations to linear ones by introducing a new variable denoted by h_{ij}^o, which is equal to $x_{ji}^o g_j^o$. Thus, the linear formulation of Constraint (A.5) will be as follows.

$$g_i^o = \sum_{j:(j,i)\in A} (h_{ji}^o + d_{ji} x_{ji}^o) \quad \forall i \in S, \forall o \in O \tag{A.13}$$

$$h_{ji}^o \leq d_{max} x_{ji}^o \quad (j,i) \in A, \forall o \in O \tag{A.14}$$

$$h_{ji}^o \leq g_j^o \quad \forall j \in S, \forall o \in O \tag{A.15}$$

$$h_{ji}^o \geq g_j^o - (1 - x_{ji}^o) d_{max} \quad (j,i) \in A, \forall o \in O \tag{A.16}$$

$$h_{ji}^o \geq 0 \quad \forall \quad (j,i) \in A, \forall o \in O \tag{A.17}$$

As a result, the linearized arc-based model results in the following problem:

$$\min \left(\sum_{o\in O} \sum_{(j,i)\in A\setminus A'} C_{ji}^o x_{ji}^o + \sum_{o\in O} \sum_{(j,t)\in A'} C_{jt}^o x_{jt}^o + \sum_{b\in B} V_b z_b \right) \tag{A.18}$$

s.t.

Constraints (A.2)–(A.4), (A.6)–(A.12), and (A.13)–(A.17)

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